

Convenient but risky government bonds*

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Abstract

How does convenience yield interact with sovereign risk and the supply of government bonds? We propose a model of sovereign debt and default in which convenience yield arises because investors are able to pledge government bonds as collateral on financial markets. Consistent with euro area data convenience yield is large if government bonds are (i) scarce due to investors' high *collateral valuation* or (ii) safe due to a small *collateral haircut* being applied to them. Calibrating the model to the data, we demonstrate that convenience yield improves the fit of sovereign default models to developed economy bond market data, contributes substantially to the public debt-to-GDP ratio, and rationalises prolonged periods of negative bond spreads – even in the presence of default risk. A large debt elasticity of investors' collateral valuation is key to these results. In this setting, highly debt-elastic collateral haircuts exacerbate collateral scarcity in crisis times, raising government bond prices and eroding fiscal discipline.

Keywords: Sovereign risk, convenience yield, collateral haircuts, public debt management

JEL classification: G12, G15, H63

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1 Introduction

Government bonds play a special role in the financial systems of developed economies, as they provide an exceptional degree of liquidity and safety to investors. This makes government bonds the asset class that is most frequently pledged as collateral on financial markets, in particular on the repo market. It also implies that investors pay a price markup for government bonds due to the collateral service that they provide (for an early reference, see [Bansal and Coleman, 1996](#)). The associated premium is usually referred to as *convenience yield*.

The literature has documented the negative relationship between convenience yield and government bond supply ([Krishnamurthy and Vissing-Jorgensen, 2012](#); [Acharya and Laarits, 2023](#)), the potential of convenience yield to reconcile the high valuations of government debt ([Jiang et al., 2022b](#); [Mian et al., 2022](#)), and its implications for public debt management ([Greenwood et al., 2015](#); [Jiang et al., 2022a](#); [Gorton and Ordoñez, 2022](#)). However, the focus so far has usually been on the United States, where sovereign risk is effectively not a concern. By contrast, several euro area countries face non-negligible default risk but are still able to issue bonds with sizable convenience yields ([Jiang et al., 2021](#)). Yet, there is limited knowledge on the interaction between convenience yield, public debt management and sovereign risk. Our paper aims to fill this gap.

Using bond market data from Italy, we document that the convenience yield of a risky government bond is driven by two distinct channels. We consider the CDS-bond basis of Italian over German government bonds, which is defined as the yield spread between German and Italian government bonds, minus Italy's credit default swap (CDS) spread. The German Bund is an alternative security that investors can use as collateral but is free of default risk ([De Fiore et al., 2023](#)), while CDS spreads are a commonly used measure of sovereign default risk. If the yield spread between Germany and Italy is larger in absolute terms than Italy's CDS spread, Italian bonds pay a higher yield than what can be explained by default risk alone. The CDS-bond basis measures this funding disadvantage of Italian vis-à-vis German bonds.

The data show that the basis increases in the scarcity of Italian bonds, as measured by the debt-to-GDP ratio. This *collateral valuation effect* has been documented and studied by [Krishnamurthy and Vissing-Jorgensen \(2012\)](#) and [Greenwood et al. \(2015\)](#) for the United States. In addition, the basis declines in Italy's default risk, as measured by the CDS spread. This is consistent with the observation that bonds pledged in financial market transactions as collateral are subject to collateral haircuts that increase in a bond's default risk. Such haircuts reduce the bond's convenience yield ([Nguyen, 2020](#) or [Nissinen and Sihvonen, 2024](#)), since investors can only pledge one minus the collateral haircut per unit of a bond. We refer to this as the *collateral haircut effect*, which has not been studied in the literature focusing on the default-free case.

Motivated by these empirical observations, our main contribution is to add convenience yield to an otherwise standard quantitative model of endogenous sovereign default. Studying the implications of convenience yield for sovereign risk and public debt management is a non-trivial task since convenience yield, sovereign risk, and the supply of government bonds are jointly and endogenously determined. To illustrate the basic interactions, we first propose a simple two-period framework featuring an impatient and risk-neutral government that issues defaultable one-period bonds to investors, who derive utility from the collateral services of government bonds. The key novel ingredients to our model relate to

investors' pricing of risky government bonds and are directly informed by the data. In the spirit of [Krishnamurthy and Vissing-Jorgensen \(2012\)](#), the collateral valuation by investors declines in available collateral. In the case of risky bonds, available collateral is given by the market value of all outstanding government debt, corrected for collateral haircuts. Haircuts in turn are linked positively to default risk, consistent with our empirical observations and with common practice on financial markets.¹

The investor problem yields a pricing schedule for government bonds that contains a convenience yield term. Convenience yield equals investors' collateral valuation, multiplied by the amount of collateral provided by the bond, i.e. one minus the collateral haircut. Consequently, the government's debt choice is affected by convenience yield in two distinct ways. First, bond prices decrease in debt issuance holding default risk constant, as collateral becomes less scarce and convenience yield declines. This is the *collateral valuation effect*. Second, holding the collateral valuation constant, default risk depresses bond prices not just by raising the default risk premia charged by investors, but also by lowering the convenience yield through the *collateral haircut effect*.

Taking as given investors' bond pricing schedule and future default incentives, the government issues bonds to maximise public spending over both periods. Absent convenience yield, optimal debt issuance equates the marginal benefit of issuing debt, given by the government's relative impatience, with the cost of doing so, given by the marginal default risk premium charged by investors ([Arellano, 2008](#)). How does convenience yield affect the government's incentive to issue debt? First, consider the special case where collateral valuation and collateral haircuts do not respond to outstanding government debt. Ceteris paribus, the resulting debt-inelastic convenience yield affects the policy trade-off by raising bond prices. This makes it more attractive for the sovereign to issue debt and take on default risk. We show analytically that a debt-inelastic convenience yield affects optimal debt issuance in the same way an increase in the government's relative impatience would do: Debt issuance and default risk increase. Debt-inelastic convenience yield thus erodes fiscal discipline.

However, if convenience yield depends on outstanding government debt, either through the collateral valuation or the collateral haircut effect, optimal debt issuance and default risk are also determined by the *marginal convenience yield*. Consider a second special case, in which investors' collateral valuation is still constant, but the collateral haircut effect is active. In this case, investors require compensation for default risk *and* the loss of collateral services through higher collateral haircuts. Consequently, the bond price schedule is more sensitive to debt issuance and the point at which the *marginal* default risk compensation equals relative impatience is reached at a lower debt level. If the *marginal* convenience yield is negative, it provides incentives for fiscal discipline.²

In the general case that now also involves a debt-elastic collateral valuation, we uncover an important complementary between the collateral valuation and collateral haircut effect. Here, debt issuance can even reduce available collateral if haircuts are very elastic to debt issuance and default risk. Then, the *haircut-corrected* market value of government bonds can decline despite an increase in outstanding debt.

¹In practice, the convenience yield of a bond depends on its haircut on various market segments. When using the term *collateral haircut*, we refer to an average haircut over different segments. For example, [Drechsler et al. \(2016\)](#) document that the Eurosystem applied a flatter haircut schedule for its standing facilities than the haircuts set on private repo markets by large central clearing counterparties (CCP). The average haircut encompasses this heterogeneity in a tractable way. For our comparative statics exercises with respect to the steepness of haircut schedules, we do not take a stance on whether it reflects the risk management practice of CCPs or central bank collateral policies (see [Nyborg, 2017](#)).

²[Mian et al. \(2022\)](#) stress the importance of debt-elastic convenience yields in the context of the United States.

If investors' valuation of collateral services is sufficiently elastic, marginal convenience yield is positive, such that bond prices become less responsive to default risk. This drastically erodes fiscal discipline beyond the case of debt-inelastic convenience yield.

We demonstrate the quantitative relevance of convenience yield and its components for a fully-fledged quantitative model of sovereign debt and default (see [Chatterjee and Eyigungor, 2012](#); [Bocola et al., 2019](#)), with a risk-averse government facing persistent revenue shocks and smoothing public spending across states by issuing defaultable long-term bonds to investors. The quantitative model is calibrated to Italian data for the time period from 2001 to 2012. Free model parameters are set to match the mean and volatility of bond spreads over the risk-free rate, CDS spreads, and debt outstanding over the full sample. The model is able to match key moments for public debt management, such as the countercyclicality of debt issuance, and generates a highly right-skewed distribution of bond and CDS spreads.

To corroborate the validity of the investor pricing kernel, and closely aligned with our empirical analysis, we use the model to price a security that provides convenience yield but is free of default risk, corresponding to a German government bond (Bund). The Italian CDS-bond basis to the Bund is *negative* in the model, which is consistent with the data and implies that the spread between Italy and Germany is larger than the CDS spread. Furthermore, Italy's relative convenience yield declines in the Italian debt-to-GDP ratio (collateral scarcity effect) and in CDS spreads (collateral haircut effect).

The government's debt and default incentives are closely linked to the government bond spread over the risk-free rate, proxied by maturity-matched interest rate swaps that reflect investors' funding conditions. This bond spread reflects default risk and convenience yield, and it is negative in around 60% of all simulated model periods. In the data, the government bond spread over the risk-free rate is negative in the early 2000s and turns positive in 2008Q4, which prompts us to interpret the time period from 2008Q4 to 2012Q4 as the crisis sub-sample. Although the model is quite parsimonious and features only one exogenous shock process, it turns out that the model replicates the drastic widening of the CDS-bond basis between Italy and Germany, i.e. the difference between Italian and German bond yields beyond what can be explained by default risk alone. The CDS-bond basis widens from around -5 basis points to almost -50 basis points in the crisis sub-sample.³

Using our calibrated model, we demonstrate that convenience yield can account for an economically relevant part of the public debt-to-GDP ratio. Without convenience yield, the government would need to be considerably more impatient in order to match the debt-to-GDP ratio from the data. Alternatively, holding the government discount factor constant, the debt-to-GDP ratio would be around 13% lower in the model without convenience yield relative to our baseline model. This number is smaller but of a similar magnitude as the estimate by [Choi et al. \(2024\)](#), who find that US public debt would be around 30% lower in the absence of convenience yield. We demonstrate that interactions between collateral valuation and collateral haircut effects that imply a positive marginal convenience yield drive this result. If we make collateral valuation a constant, but keep the collateral haircut schedule in its baseline form (negative marginal convenience yield), or set convenience yield itself to a constant, the discount factor needs to be reduced almost as much as in the model version without convenience yield in order to

³Such a disproportional widening in spreads is conceptually distinct from a model with risk-averse investors, which can rationalise government bond spreads increasing beyond pure default risk compensation during a fiscal crisis (for a discussion, see [Lizarazo, 2013](#)), but would not be capable of generating negative bond spreads in safe times.

match the debt-to-GDP ratio. The shape of convenience yield, rather than its level, is more important for sovereign debt and default risk, which is in line with recent results by [Mian et al. \(2022\)](#) for (non-defaultable) risk-free government bonds.

The shape of investors' valuation of collateral services also has important implications for the design of collateral haircuts, which are at least to a certain degree under the control of central banks and financial market regulators. Conventional wisdom suggests that collateral haircut schedules provide incentives for fiscal discipline if they respond strongly to default risk. However, highly elastic haircut schedules increase debt outstanding and the default rate in our model: They exacerbate collateral scarcity in crisis times by drastically reducing the amount of collateral provided by each bond. For empirically plausible elasticities of collateral valuation with respect to available collateral, this implies that the marginal convenience yield is positive and, thus, erodes fiscal discipline.

Reassuringly, our main policy-relevant finding that more elastic collateral haircut schedules erode fiscal discipline is robust to extending the model with additional features that may confound our results. First, we consider collateral haircuts that depend on the one-year-ahead instead of the one-quarter-ahead default probability. Second, we allow for trading frictions that give rise to market illiquidity and elevated bid-ask spreads in crisis periods. In a third and fourth extension, we allow investors' collateral valuation and the supply of alternative collateral to increase during crisis times, respectively. This might alter sovereign borrowing incentives in high-risk periods. Fifth, we allow for risk-averse investors, such that debt issuance and default impact bond pricing by affecting investor consumption. Finally, we also provide re-calibrations of the baseline model to data from Spain and Portugal.

In all extensions and re-calibrations, our key comparative statics remain intact. More elastic collateral haircut schedules fail to provide fiscal discipline incentives and the government runs a higher debt-to-GDP ratio on average. As this also implies a higher default rate, government welfare declines. Despite this and contrary to conventional wisdom, collateral availability also contracts, as the larger default rate implies higher average haircuts and a larger variation of collateral availability over time, which in turn is detrimental to investor welfare. Our results cast doubt on the notion of a clear trade-off between increasing collateral availability to investors and maintaining fiscal discipline incentives, once the valuation of collateral services and the supply of collateral are fully endogenous.

Related literature. Our paper draws on the quantitative sovereign default literature, which mostly focuses on external public debt and emerging economies. [Aguiar et al. \(2016\)](#) provide an extensive survey. In [Sosa-Padilla \(2018\)](#) and [Perez \(2018\)](#), government bonds provide public liquidity to domestic agents, which yields additional repayment incentives. Similar to [Bocola et al. \(2019\)](#), we reinterpret the established sovereign default framework à la [Eaton and Gersovitz \(1981\)](#) and [Arellano \(2008\)](#) as a relationship between a government and lenders that abstracts from lenders' place of residence, and apply this framework to a developed economy. Specifically, [Bocola et al. \(2019\)](#) show that total government debt, rather than external government debt, is the relevant determinant of default risk of developed economies. Complementary to our paper, [Lengwilder and Orphanides \(2023\)](#) study the interactions between collateral frameworks in the Eurosystem and the liquidity of government bonds in a model that has multiple equilibria. For a quantitative analysis of self-fulfilling debt crisis in the euro area and the ECB's role in coordinating on the good equilibrium through OMT announcements, see also [Bocola and Dovis \(2019\)](#).

Our paper is also related to recent studies that emphasise the role of convenience yield for optimal public debt management. [Canzoneri et al. \(2016\)](#) study optimal monetary and fiscal policy under commitment when government bonds provide transaction services, leading to a trade-off between tax smoothing and liquidity provision (see also [Angeletos et al., 2023](#)). [Jiang et al. \(2022a\)](#) analyse convenience yield in a model in which the government trades off insuring bondholders and taxpayers. [Greenwood et al. \(2015\)](#) study optimal debt maturity management when short-term government bonds are valued by investors for having money-like properties. [Brunnermeier et al. \(2024\)](#) focus on the role of sovereign debt as safe asset that enables households to self-insure against aggregate risk, which generates cyclical variation in convenience yields. [Mian et al. \(2022\)](#) explicitly consider the debt-elasticity of convenience yield in a model of optimal fiscal policy. These papers consider the interaction between public debt and the convenience yield of government bonds in the context of (optimal) fiscal policy. By contrast, [Bonam \(2020\)](#) studies the implications of convenience yield in a New Keynesian model with fiscal policy governed by an exogenous policy rules. [Fu et al. \(2023\)](#) and [Cieslak et al. \(2023\)](#) document the interactions of inflation (expectations), government bond supply and convenience yield and study its implications through the lenses of New Keynesian models. [Arcidiacono et al. \(2024\)](#) study convenience yield spillovers between different safe euro area countries. Due to its focus on the United States or alternative safe asset providers such as Japan or Germany, this literature abstracts from sovereign risk.

Layout. The paper is structured as follows. Section 2 provides motivating empirical evidence on the relationship between convenience yield, default risk, and government bond supply. In Section 3, we introduce a simple two-period model of convenience yield and sovereign risk. Section 4 presents a quantitative infinite-horizon model, which we calibrate in Section 5. Section 6 explores the interaction between convenience yield, credit risk, and government bond supply. Section 7 concludes.

2 Convenience yield and default risk in the data

This section presents motivating empirical evidence on the presence of convenience yield on risky government bonds. We first provide some important definitions concerning government bond yields and convenience yield and discuss how we relate them to our structural model of convenience yield and sovereign default. Using Italian government bond data from 2001Q1 to 2012Q4, we then show that, *ceteris paribus*, our measure of convenience yield declines in the supply of (Italian) government bonds and in their default risk. By choosing this sample, we exclude periods with large scale asset purchases by the Eurosystem, which are difficult to address in our model framework and in the data. A complete list of the data sources can be found in Table A.1. We use these empirical observations when calibrating our structural model to the data in Section 5.

Definitions. We first describe how we measure convenience yield in the data. As is customary in the empirical literature, we define convenience yield as a residual of the government bond yield r_t , a replication portfolio of the risk-free rate r_t^{rf} and a measure of sovereign default risk. We use sovereign CDS spreads, denoted by cds_t , with the same maturity as a measure of default risk and maturity-matched interest rate swaps as a measure of investor funding conditions r_t^{rf} . In a canonical sovereign default model ([Eaton and Gersovitz, 1981](#)), these three rates are linked by the no arbitrage relationship $r_t = r_t^{rf} +$

cds_t . The empirical literature has clearly rejected this relationship. Following [Gârleanu and Pedersen \(2011\)](#), [Krishnamurthy and Vissing-Jorgensen \(2012\)](#) and [Gorton and Ordoñez \(2022\)](#), our structural model ties the valuation of this no arbitrage relationship to the collateral services that government bonds provide. We denote investors' valuation of collateral services by cv_t in the following, which we assume to be common to all (euro area) bonds.

To see how investors' collateral valuation affects the pricing of government bonds, it is useful to start with the case of risk-free government bonds, on which the literature has focused up to this point. The data show that the yields of risk-free sovereign bonds, such as German Bunds, are persistently smaller than common measures of investor funding conditions. Since German Bunds are approximately free of default risk ($cds_t^{DE} \approx 0$), the yield of German government bonds satisfies $r_t^{DE} < r_t^{rf}$. However, consistent with the notion that risky government bonds also provide convenience yield, we also observe $r_t^{IT} < r_t^{rf} + cds_t^{IT}$ for each quarter in the full sample, i.e. Italy is able to borrow at a lower rate than what would be justified by default risk compensation (cds_t) and investor funding cost (r_t^{rf}). We refer to this discrepancy as convenience yield in the following, i.e.

$$cy_t \equiv r_t^{rf} + cds_t^{IT} - r_t^{IT} . \quad (1)$$

Its full sample average amounts to plus 30 basis points. This measure of convenience yield varies with sovereign risk and is also affected by time-variation in investors' collateral valuation cv_t . We need to isolate collateral demand components from the interactions between default risk, government bond supply and the convenience yield on Italian government bonds in order to inform our structural model. To do so, we follow the literature and relate Italy's convenience yield to the convenience yield of risk-free benchmark bonds, see [Du et al. \(2018\)](#). It comes natural to use the German convenience yield $cy_t^{DE} \equiv r_t^{rf} + cds_t^{DE} - r_t^{DE}$ (see also [Jiang et al., 2021](#)) as the benchmark. We define the CDS-bond basis as

$$basis_t^{DE} \equiv cy_t - cy_t^{DE} = (cds_t^{IT} - r_t^{IT}) - (cds_t^{DE} - r_t^{DE}) = (r_t^{DE} - r_t^{IT}) - (cgs_t^{DE} - cds_t^{IT}) . \quad (2)$$

A negative $basis_t^{DE}$ can be interpreted as funding disadvantage of Italian government bonds relative to a risk-free government bond. If the spread between Italian and German government bonds $s_t^{IT,DE} \equiv r_t^{IT} - r_t^{DE}$ exceeds the CDS differential ($cgs_t^{IT} - cds_t^{DE}$) in absolute terms, investors require a larger yield for holding Italian government bonds than what is justified by default compensation alone. By linking convenience yield to collateral services, our model attributes the excessive funding disadvantage of Italian government bonds – beyond what can be explained by sovereign risk alone – to their large collateral haircuts.

By contrast, German government bonds could be pledged as collateral at very low collateral haircuts throughout the European debt crisis ($haircut_t^{DE} \approx 0$), such that the convenience yield on Bunds equals investors' collateral valuation $cy_t^{DE} = cv_t$. Consequently, we can express the Italian convenience yield as $cy_t \approx (1 - haircut_t^{IT}) \cdot cv_t$. If German bonds are approximately default risk-free ($cgs_t^{DE} \approx 0$), we can use (1) to approximate Italy's spread over Bunds as

$$s_t^{IT,DE} = r_t^{IT} - r_t^{DE} \approx r_t^{rf} + cds_t^{IT} - (1 - haircut_t^{IT}) \times cv_t - (r_t^{rf} - cv_t) = cds_t^{IT} + haircut_t^{IT} \times cv_t \quad (3)$$

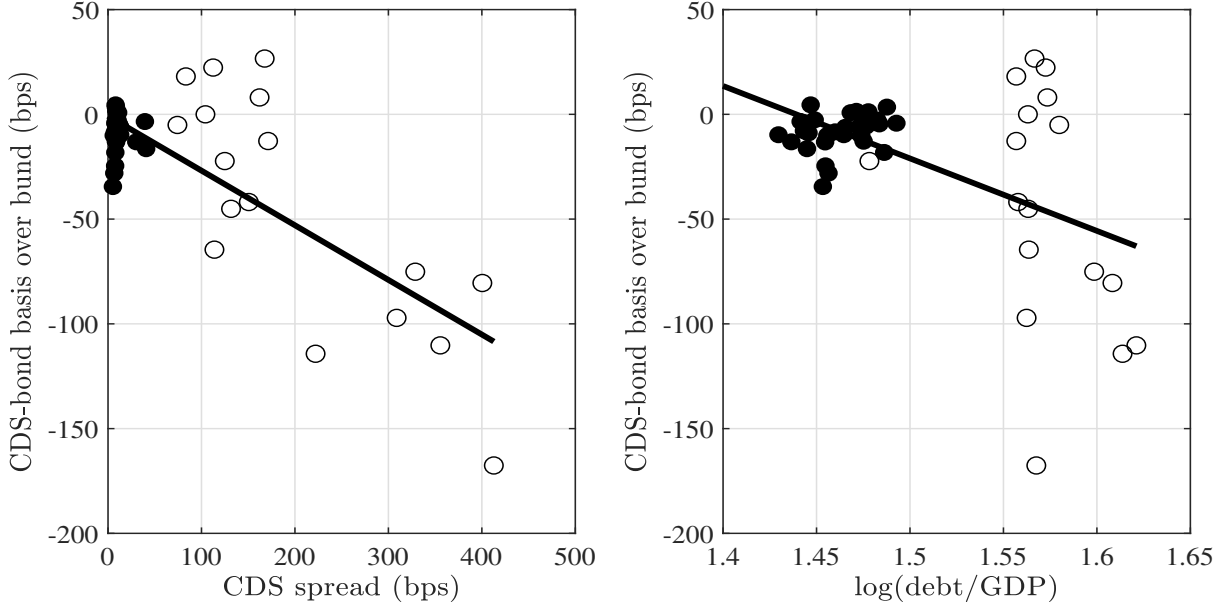


Figure 1: Relationship between convenience yield and default risk. The debt-to-GDP ratio is defined with respect to quarterly GDP. Convenience yield and CDS spreads are in basis points. The pre-crisis sub-sample is indicated by black dots, the post-crisis sub-sample by white dots. The correlation coefficients are -0.78 and -0.52, respectively, and are highly significant.

As long as investors' demand for collateral is not satiated, Italy's spread over the Bund will exceed its CDS spread. Using (3) in equation (2), we obtain a simple expression for the relative convenience yield of Italy vis-à-vis Germany,

$$basis_t^{DE} = cds_t^{IT} - s_t^{IT,DE} = -haircut_t^{IT} \times cv_t, \quad (4)$$

which will be negative on average and large in absolute terms when Italian default risk and its collateral haircuts are high. Further empirical evidence on the relationship between collateral haircuts and government bond yields is provided by [Nguyen \(2020\)](#) as [Nissinen and Sihvonon \(2024\)](#).

Results The left panel of Figure 1 shows how $basis_t^{DE}$ on the y-axis is linked to Italian CDS spreads on the x-axis. Solid black points refer to the pre-crisis sub-sample, which is characterized by low default risk and limited variation in Italy's funding disadvantage. The right panel shows how $basis_t^{DE}$ is linked to the supply of Italian government bonds. The pre-crisis sub-sample is also associated with relatively low debt-to-GDP ratios, but still features a negative correlation between bond supply and Italy's funding conditions, relative to a risk-free bond. In the crisis sub-sample, the funding disadvantage widens to more than 100 basis points in absolute terms at the peak of the sovereign debt crisis, which is also associated with a considerable increase in the debt-to-GDP ratio. In Appendix A, we show that a very similar picture emerges for Spain, Portugal, Greece and Ireland, which paints a conclusive picture of the relationship between convenience yield, sovereign risk and government bond supply. Guided by the empirical relationship between default risk, government bond supply and convenience yield, we now introduce a simple analytical framework that allows us to study the implications of convenience yield for sovereign risk and public debt management in a transparent way.

3 A simple model of default risk, convenience yield, and haircuts

In this section, we illustrate the basic interactions between sovereign risk, convenience yield and debt issuance in a two-period setting. We show that, if convenience yield is debt-inelastic, the government issues more bonds and defaults more frequently than it would in the absence of convenience yield. Introducing a debt-elastic convenience yield potentially overturns this result: Convenience yield can provide fiscal discipline. We then provide a decomposition of the debt-elasticity of convenience yield into the debt-elasticity of its components (collateral valuation and collateral haircuts). Since the debt-elasticity is to some degree under the control of central banks and financial regulators, we illustrate the effects of changing the debt-elasticity of haircuts on the availability of collateral and on sovereign risk.

Environment. We consider an optimal debt issuance problem of an impatient and risk-neutral government. Its discount factor is denoted by $\beta < 1$. At time $t = 0$, it issues bonds b at price $q(b)$ to a mass-one continuum of competitive investors who do not discount the future. At date $t = 1$, the government draws a revenue shock τ with cumulative density function $F(\tau)$ and defaults if debt outstanding exceeds government revenues. The default probability associated with the debt choice in $t = 0$ is therefore simply given by $F(b)$. In the event of default, there is no debt recovery for investors and the government does not consume anything.

We furthermore assume that investors are risk-neutral and have deep pockets in the sense that their consumption is independent of a potential sovereign default and that they are always willing to lend to the government. This assumption is in line with [Bocola et al. \(2019\)](#), who show that the public debt-to-GDP ratio is relevant in the context of developed economy sovereign default if domestic and foreign investors can trade government bonds freely. This is arguably the case for euro area countries. Following this argument, the distinction between a small and a large open economy is less relevant for our main results. We drop the risk-neutral pricing of bond payoffs in an extension for our quantitative model analysis. Here, investors' valuation of pecuniary benefits also depends on the actions of the sovereign. This does not change the comparative statics with respect to convenience yield.

Investors value government debt for their pecuniary benefits, i.e. their expected future payoff, as well as their *collateral services*, such that they are willing to pay a premium on government bonds above what is justified by default risk compensation. Since only very few assets are generally accepted as collateral on financial markets, we assume that convenience yield depends on government bonds outstanding, potentially corrected for collateral haircuts. This assumption is motivated by the fact that risky government bonds are still safer and more liquid than most alternative asset classes. For example, central government debt made up around 50% of all eligible collateral for refinancing operations with the Eurosystem in the run-up and during the European debt crisis. [Mancini et al. \(2015\)](#) document that government bonds make up a considerably larger share of eligible collateral on the private repo market. In our model, collateral services are the only reason why investors' pricing of government bonds differs from the discounted expected payoff and we abstract from other special features of government bonds, such as their size.

Based on the empirical observations presented in Section 2, we assume that *collateral valuation* is captured by function $u(\theta)$, which satisfies $u'(\cdot), -u''(\cdot) \geq 0$, where the bonds' collateral value corresponds to the *haircut-adjusted* expected payoff $\theta \equiv (1 - \kappa(b))(1 - F(b))b$. Importantly, θ is the relevant

argument regarding the utility that investors gain from collateral services. We adopt the timing assumption that investors value collateral services at the beginning of period $t = 1$, before tax revenues are realised and the government might default. The functional form for investors' collateral valuation can be micro-founded based on uninsurable liquidity shocks that investors can settle either cheaply by borrowing against collateral or by using more expensive unsecured borrowing, see [Bianchi and Bigio \(2022\)](#) or [De Fiore et al. \(2023\)](#) and references therein. The amount that investors can borrow at favourable conditions on the secured market segment is limited to the market value of government bonds, corrected for collateral haircuts. Consequently, investors' collateral valuation declines in haircuts.⁴

Lastly, we assume that neither the consumption of investors nor their loss of collateral services in a default event enters the government's objective, either because they reside outside the economy or because domestic agents can trade with foreign investors on international financial markets in a frictionless manner. Under this assumption, the total supply of government bonds – and not the economy's external debt stock – becomes the relevant state variable that determines sovereign risk. For a detailed discussion, we refer to [Bocola et al. \(2019\)](#). Furthermore, consistent with the literature, public debt outstanding affects the government's supply of public goods in the repayment case, but the value of the outside option (sovereign default) is independent of public debt outstanding ([Arellano, 2008](#)).

The presence of collateral valuation $u(\cdot)$ gives rise to a convenience yield $\Lambda(b)$ in investors' government bond pricing condition.⁵ This condition links investors' willingness to pay for government bonds to outstanding debt:

$$q(b) = (1 - F(b))(1 + \Lambda(b)) . \quad (5)$$

Convenience yield $\Lambda(b) = (1 - \kappa(b))u'(\theta)$ depends on the pure collateral valuation component $u'(\theta)$ and the collateral haircut component $1 - \kappa(b)$. In our simple model, we can assume that haircuts are increasing in debt issuance $\frac{d\kappa(b)}{db} > 0$. In the quantitative model analysed later, haircuts instead are positively related to default risk. In the two-period setting, default risk $F(b)$ is immediately determined by debt issuance b , such that we can conveniently express the dependence of collateral haircuts in terms of its debt choice.

Debt issuance. Taking as given investors' bond pricing condition (5), the government maximises

$$\max_b \quad q(b)b + \beta \int_b^\infty (\tau - b)dF(\tau) .$$

Throughout this section, we will refer to the relationship between debt b and bond revenues raised in that period, $q(b)b$, as the *debt issuance Laffer curve* ([Arellano, 2008](#)). The first-order condition requires that the government issues debt up to the point at which expected repayment obligations (per unit of

⁴Our specification of the investor pricing kernel conceptually different from a model of preferred habitat investors that demand government bonds issued by a particular country. The valuation of holding such bonds would solely depend on the market value of government bonds outstanding and would not depend on the haircut component. [Moretti et al. \(2023\)](#) augment an emerging market sovereign default model by an investor pricing kernel derived from a demand-system asset pricing condition. Consistent with our theory, they find that such a pricing kernel provides fiscal discipline.

⁵We derive the bond pricing condition from the maximisation problem of investors who value collateral services, which is conceptually related to the intermediary asset pricing literature, see for example [He and Krishnamurthy \(2013\)](#). Different from this literature, the bond pricing condition in our paper reflects investors' demand for liquid assets rather than investors' equity constraints.

debt b) equal the slope of the debt issuance Laffer curve:

$$\beta(1 - F(b)) = q(b) + \frac{dq(b)}{db}b. \quad (6)$$

Without convenience yield ($\Lambda = 0$), the bond price simply reflects expected repayment $q(b) = 1 - F(b)$, and its derivative is given by $\frac{dq(b)}{db} = f(b) \equiv \frac{dF(b)}{db}$. Let the hazard rate of government revenues τ be denoted as $h(\tau) \equiv \frac{f(\tau)}{1 - F(\tau)}$. Plugging this definition, together with the bond pricing expression, into the first-order condition (6), optimal debt issuance absent convenience yield, denoted as b^0 , is implicitly defined through

$$h(b^0)b^0 = 1 - \beta. \quad (7)$$

The left-hand side, $h(b^0)b^0$, can be interpreted as the elasticity of default losses with respect to debt issuance. Assuming monotonicity of the hazard rate $h(\cdot)$, which is satisfied by the log-normal distribution, the debt choice b^0 and the associated default probability is pinned down uniquely.

Adding convenience yield. The bond price schedule (5) with convenience yield is given by $q(b) = (1 - F(b))(1 + (1 - \kappa(b))u'(\theta))$, while its derivative reads $\frac{dq(b)}{db} = -(1 + \Lambda(b))f(b) + (1 - F(b))\frac{d\Lambda(b)}{db}$. Both objects, the bond price and its derivative, are larger in absolute terms than in the case without convenience yield. This implies that, in the risky borrowing region, each additional unit of debt (*i*) increases public consumption in the first period to a larger degree than in the absence of convenience yield, and (*ii*) has a more pronounced negative effect on the bond price. Plugging these two competing effects on $q(b)$ and $\frac{dq(b)}{db}$ into equation (6) and expressing it in terms of the hazard rate, we have

$$h(b)b = \frac{1 - \beta + \Lambda(b) + b\frac{d\Lambda}{db}}{1 + \Lambda(b)}. \quad (8)$$

While the left-hand side is still given by the elasticity of default losses with respect to debt issuance, the right-hand side now also contains a convenience yield wedge in addition to relative impatience $1 - \beta$. While $\Lambda(b)$ is positive by assumption, the sign of marginal convenience yield $\frac{d\Lambda(b)}{db}$ is not known a priori.

To pin down the effect of convenience yield on debt issuance, it is helpful to decompose the marginal convenience yield $\frac{d\Lambda(b)}{db}$ into a collateral scarcity effect and a collateral haircut effect. Specifically, the marginal convenience yield $\frac{d\Lambda(b)}{db}$ can be expressed as

$$\frac{d\Lambda(b)}{db} = \underbrace{(1 - \kappa(b))u''(\theta)\frac{d\theta(b)}{db}}_{\text{collateral scarcity effect}} - \underbrace{\frac{d\kappa(b)}{db}u'(\theta)}_{\text{collateral haircut effect}}. \quad (9)$$

While the collateral haircut effect is negative by construction, the collateral scarcity effect can be positive or negative.

In a model with risk-free debt, i.e. $\kappa(b) = 0$, the collateral haircut effect vanishes by construction. With respect to the collateral scarcity effect, note that the effect of debt issuance on collateral reduces to $\frac{d\theta(b)}{db} = \frac{dbq(b)}{db} > 0$, which is positive since it is not optimal to choose a debt level on the downward-sloping part of the *debt issuance Laffer curve*. Furthermore, we have assumed that $u''(\theta) < 0$, such

that more debt issuance makes collateral less scarce and reduces convenience yield. In the model with default risk, debt issuance can reduce the haircut-corrected market value of government bonds if haircuts are very sensitive to debt issuance. Formally, the effect of debt issuance on available collateral is given by

$$\frac{d\theta(b)}{db} = (1 - F(b)) \left[\underbrace{(1 - \kappa(b))(1 - h(b)b)}_{\text{ambiguous sign}} + \underbrace{\frac{d(1 - \kappa(b))}{db}b}_{<0} \right]. \quad (10)$$

The haircut component declines in debt outstanding by assumption. In addition, the market value of bonds declines rapidly in the risky borrowing region where $h(b)b$ is large, see also the *debt issuance Laffer curve* in the left panel of Figure 2. Issuing an additional unit of bonds can decrease the total amount of collateral available to investors, which then increases their collateral valuation.

To relate this potential ambiguity to the debt choice, we use the observation that marginal convenience yield $\frac{d\Lambda(b)}{db}$ only enters the first-order condition (8) together with debt issuance b . Exploiting this, equation (11) expresses the optimal debt choice with debt-elastic convenience yield in closed form,

$$h(b^*)b^* = \frac{1 - \beta + \Lambda(b^*)(1 + \varepsilon_{\bar{\kappa}}(b^*) - R(\theta^*)) - R(\theta^*)\varepsilon_{\bar{\kappa}}(b^*)}{1 + \Lambda(b^*)(1 - R(\theta^*))}, \quad (11)$$

where $\theta^* = (1 - \kappa(b^*))(1 - F(b^*))b^*$ is the haircut-corrected value of debt outstanding, $R(\theta) \equiv \frac{-\theta u''(\theta)}{u'(\theta)}$ is the elasticity of investors' marginal utility from collateral holdings and $\varepsilon_{\bar{\kappa}}(b) \equiv \frac{b \frac{d(1 - \kappa(b))}{db}}{(1 - \kappa(b))}$ is the elasticity of the haircut component $1 - \kappa(b)$ with respect to debt issuance.⁶

Equation (11) shows that the elasticity of convenience yield affects the government's optimal debt choice through two distinct channels: First, through the curvature of the collateral valuation function $u(\cdot)$ and, second, through the curvature of the collateral haircut function $\kappa(b)$. To provide more intuition for this decomposition and its relation to the micro-foundations of both channels, it is useful to consider several special cases of equation (11).

⁶We can rewrite marginal convenience yield by multiplying equation (9) with b on both sides:

$$b \frac{d\Lambda(b)}{db} = b(1 - \kappa(b))u''(\theta) \frac{d\theta}{db} + b \frac{d(1 - \kappa(b))}{db} u'(\theta),$$

and plug in the derivative of total collateral with respect to debt issuance Equation (10) to get:

$$b \frac{d\Lambda(b)}{db} = b(1 - \kappa(b))u''(\theta)(1 - F(b)) \left[(1 - \kappa(b))(1 - h(b)b) + \frac{d(1 - \kappa(b))}{db}b \right] + b \frac{d(1 - \kappa(b))}{db} u'(\theta).$$

We can factor out $(1 - \kappa(b))u'(\theta)$ and use the definitions of $\theta = (1 - \kappa(b))(1 - F(b))b$ and $\varepsilon_{\bar{\kappa}}(b) = \frac{b \frac{d(1 - \kappa(b))}{db}}{(1 - \kappa(b))}$ to write

$$b \frac{d\Lambda(b)}{db} = (1 - \kappa(b))u'(\theta) \times \left\{ \frac{u''(\theta)}{u'(\theta)} \theta \left((1 - h(b)b) + \varepsilon_{\bar{\kappa}}(b) \right) + \varepsilon_{\bar{\kappa}}(b) \right\}.$$

Together with the definitions of convenience yield $\Lambda(b) = (1 - \kappa(b))u'(\theta)$ and of the elasticity of marginal utility $R(\theta) = \frac{-\theta u''(\theta)}{u'(\theta)}$, we can plug this into the first-order condition (8) to obtain

$$h(b^*)b^* = \frac{1 + \Lambda(b^*) + \Lambda(b^*) \times \left\{ -R(\theta^*) \left((1 - h(b^*)b^*) + \varepsilon_{\bar{\kappa}}(b^*) \right) + \varepsilon_{\bar{\kappa}}(b^*) \right\} - \beta}{1 + \Lambda(b^*)}.$$

Rearranging for $h(b^*)b^*$ yields equation (11).

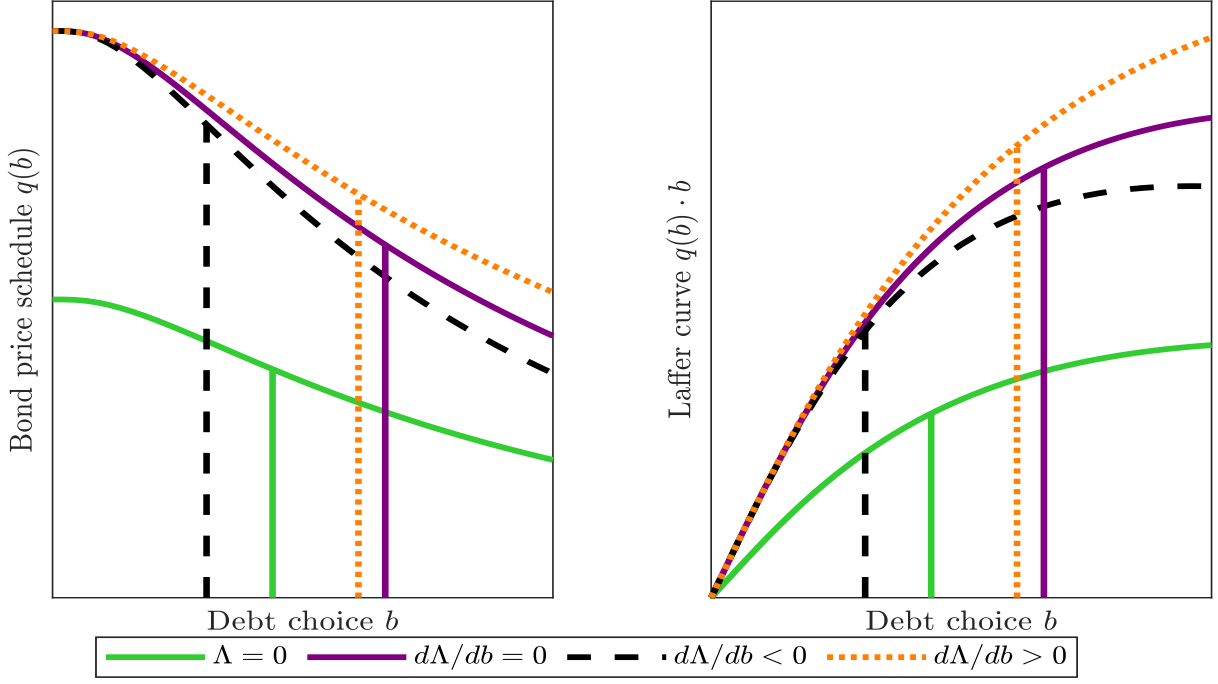


Figure 2: The effect of convenience yield on debt issuance

Special case I: Constant convenience yield If the convenience yield is a constant $\tilde{\Lambda}$, the debt choice (8) reduces to

$$h(\tilde{b})\tilde{b} = \frac{1 - \beta + \tilde{\Lambda}}{1 + \tilde{\Lambda}}. \quad (12)$$

In this special case, convenience yield increases the incentives to issue debt and take on default risk. Formally, the right hand side increases in $\tilde{\Lambda}$ and the monotone hazard rate assumption on government revenues on the left hand side implies that $\tilde{b} > b^0$. The case with constant convenience yield is reflected by the solid purple line in Figure 2. The solid green line reflects the case without convenience yield. The left panel shows the bond price schedule that maps any potential debt choice by the sovereign into investors' willingness to pay for government bonds. For any b , the bond price is larger whenever there is a positive convenience yield.

The right panel of Figure 2 depicts the government's *debt issuance Laffer curve*, i.e. the amount of resources $q(b)b$ raised by issuing b units of debt. The *Laffer curve* is also located above the baseline case for any positive convenience yield. Note that this is not informative about the government's debt choice, which equates the slope of the *Laffer curve* with the borrowing motive, i.e. relative impatience. Graphically, a debt-inelastic convenience yield shifts this point to the right, such that the optimal debt level increases, which is indicated by the vertical purple and green lines, respectively. In this simplified setting, a larger debt choice is associated with a higher default probability $F(b)$, but not necessarily a lower bond price, as shown in the left panel.

Special case II: Adding debt-elastic collateral haircuts As a next step, we allow convenience yield to depend on default risk though collateral haircuts, but we keep investors' valuation of collateral ser-

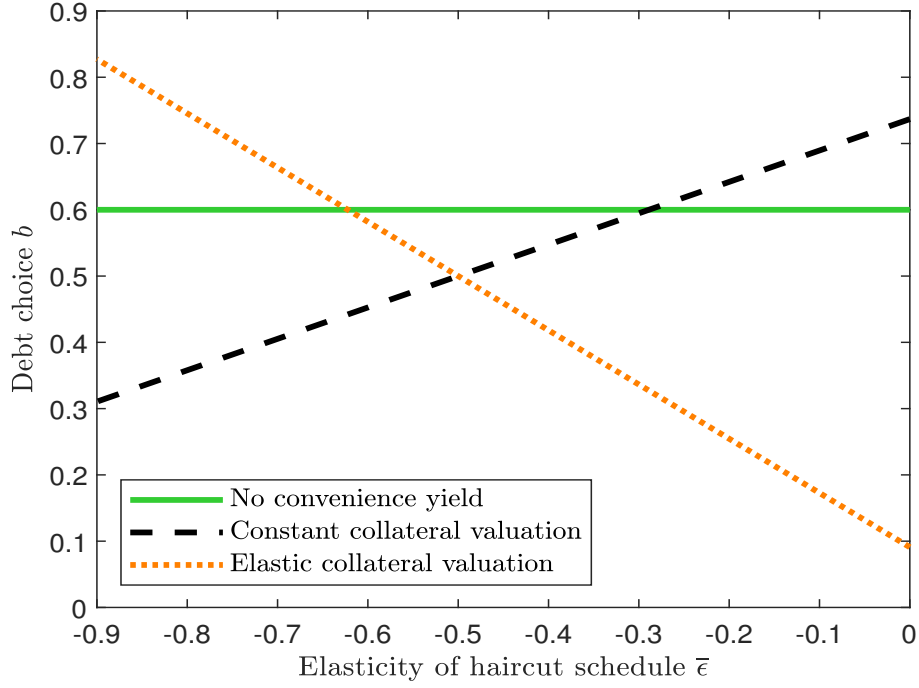


Figure 3: The effect of haircut elasticities on debt issuance

vices constant. Note that the elasticity of the collateral haircut component $1 - \kappa(b)$ with respect to debt issuance $\varepsilon_{\bar{\kappa}}(\bar{b})$ is negative by the assumption that collateral haircuts increase in debt issuance via sovereign risk. For simplicity, we assume that the elasticity of the collateral valuation component is constant, i.e. $\varepsilon_{\bar{\kappa}}(b) = \bar{\varepsilon} < 0$. Then, for a debt-inelastic collateral valuation component, the debt choice (11) simplifies to

$$h(\bar{b})\bar{b} = \frac{1 - \beta + \Lambda(\bar{b})(1 + \bar{\varepsilon})}{1 + \Lambda(\bar{b})}. \quad (13)$$

Equation (13) shows that debt-elastic collateral haircuts provide disciplinary incentives to the government: By reducing the convenience yield wedge in the denominator, debt-elastic haircuts counteract the basic positive effect of convenience yield on debt issuance described in equation (12).

Formally, the debt-elastic haircut component ensures that marginal convenience yield $\frac{d\Lambda}{db}$ is negative. The negative marginal convenience yield directly enters the bond price schedule and, thereby, the government's untransformed first order condition (8). This case is represented by the dotted black line in Figure 2. The left panel illustrates that the bond price schedule declines more quickly in the government's debt choice than in the case with constant convenience yield. The same pattern can be observed for the *Laffer curve* in the right panel. Consequently, the optimal debt choice is smaller than in the case with constant convenience yield, i.e. $\bar{b} < \tilde{b}$.

Special case III: Adding debt-elastic collateral valuation As a final step, we introduce a debt-elastic collateral valuation, i.e. $u'(\cdot) > 0$. For simplicity, we set the elasticity of investors' marginal collateral valuation to a constant $R(\theta^*) = \hat{R}$. The debt-inelastic case is nested as the special case $\bar{R} = 0$. We maintain the assumption of a constant elasticity of the collateral haircut component with respect to debt

issuance ($\varepsilon_{\bar{K}}(b) = \bar{\varepsilon} < 0$) and can simplify (11) to

$$h(\hat{b})\hat{b} = \frac{1 - \beta + \Lambda(\hat{b})(1 - \hat{R})(1 + \bar{\varepsilon})}{1 + \Lambda(\hat{b})(1 - \hat{R})}. \quad (14)$$

If collateral haircuts would not respond to debt issuance ($\bar{\varepsilon} = 0$), the collateral scarcity channel provides disciplining incentives to the government, i.e. $\hat{b} < \tilde{b}$. For sufficiently elastic collateral valuation effects, i.e. $\hat{R} > 1$, the right hand side of equation (14) implies that the optimal debt choice is again lower than in the case without convenience yield. Similar to the special case with debt-elastic haircuts, marginal convenience yield becomes negative in this case $\frac{d\Lambda}{db}$, which is also reflected by the dotted black line in Figure 2.

The key insight of our model obtains if collateral haircuts and collateral valuation both respond to debt issuance, i.e. $\bar{\varepsilon} < 0$ and $\hat{R} > 0$. In isolation, each component of convenience yield provides disciplinary incentives to the government. However, their interaction term gives rise to non-monotonic interactions, since the government strategically takes into account collateral scarcity effects on the pricing of its bonds. This can make the collateral scarcity and collateral haircut effects *complements*, as they can jointly turn the sign of marginal convenience yield positive.

Consider first the case where collateral valuation is relatively inelastic, i.e. $\hat{R} < 1$, a more elastic haircut component provides discipline incentives to the government. When $\bar{\varepsilon}$ is sufficiently large in absolute terms, the optimal debt issuance can be even smaller than in the case without convenience yield, see the solid green line in Figure 3. The dotted black line in Figure 3 summarises how the optimal debt choice varies with the debt-elasticity of the collateral haircut component. For very negative $\bar{\varepsilon}$, convenience yield responds strongly to additional debt issuance. The government takes this into account and optimally chooses a debt level even below the solid green line.

In the second case, collateral valuation itself provides incentives to reduce debt issuance ($\hat{R} > 1$), which holds even with constant collateral haircuts ($\bar{\varepsilon} = 0$). In this case, there is a *complementarity* between both components and a greater haircut elasticity increases debt issuance and default risk. The reason behind this is that haircuts also affect collateral valuation indirectly through their effect on θ , see Equation (10). If larger haircuts induce a sufficiently great reduction in available collateral θ and if this reduction at the same time induces a sufficiently large collateral scarcity effect ($u''(\theta)$ large in absolute terms), marginal convenience yield $\frac{d\Lambda}{db}$ is positive and bond prices can increase locally in response to debt issuance. This situation is consistent with the positive effects that the negative rating outlook had on US borrowing conditions in 2011, when the arrival of bad news about the sustainability of US public debt decreased US government bond yields.

Our simple model abstracted from consumption smoothing motives by the government and from debt dilution, which are important determinants of sovereign borrowing and default incentives, see for example Hatchondo et al. (2016) or Aguiar and Amador (2020). Therefore, we embed convenience yield into a setting with long-term debt, strategic sovereign default, and positive debt recovery in the next section. We use this framework to assess whether convenience yield itself and the non-monotonic interaction of collateral valuation and haircut components have a quantitatively relevant impact on sovereign debt and default risk.

4 Quantitative model of convenience yield and endogenous default

Time is discrete and denoted as $t = 0, 1, 2, \dots, \infty$. The government receives random tax revenues, cares about the supply of a public good, and can trade long-term government bonds with competitive investors. Following the quantitative sovereign default literature, the government cannot commit to future repayment and debt issuance (see [Arellano, 2008](#)). First, we show the investor problem, which yields a pricing schedule for government bonds. Then, we describe the government's policy problem, taking as given the pricing schedule, which, together with bond market clearing, characterises the equilibrium of the model.

Investors. Investors have deep pockets and discount consumption at the exogenously given, time-invariant rate $r^{rf} > 0$. They maximise their expected life-time utility $\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \frac{c_t + u(\theta_t)}{(1+r^{rf})^t} \right]$, where c_t denotes an investor's consumption and $u(\theta_t)$ is the convenience benefit of holding government bonds. The collateral value of government bonds $\theta_t \equiv (1 - \kappa_t)m_t b_{t-1}$ consists of three parts. The first part is the stock of government bonds b_{t-1} held at the beginning of period t . The second part is given by the expected bond payoff $m_t \equiv \mathbb{E}_{t-1}[k_t]$, with k_t being the realised bond payoff in period t , which crucially depends on the government's decision to default.⁷ The third part is the collateral haircut κ_t imposed on government bonds, which depends on the credit risk of the bond (see further below). Letting the collateral value θ_t depend on the expected bond payoff is consistent with mark-to-market practices on repo and securities lending markets. The first-order condition of the investor problem yields the bond pricing equation

$$q_t = \frac{1}{1+r^{rf}} \times \underbrace{m_{t+1}}_{\text{expected payoff}} \times \underbrace{(1 + \Lambda_{t+1})}_{\text{convenience yield}}, \quad (15)$$

where convenience yield consists of two components, associated with haircuts and collateral valuation, respectively:

$$\Lambda_{t+1} \equiv \underbrace{(1 - \kappa_{t+1})}_{1-\text{collateral haircut}} \times \underbrace{u'(\theta_{t+1})}_{\text{collateral valuation}}. \quad (16)$$

Government. Each period, the government receives exogenous revenues τy_t , where τ is a constant income tax rate and y_t is random domestic income, supplies the public good g_t , issues long-term bonds and decides whether to repay its debt to maximise

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t v(g_t) \right], \quad (17)$$

⁷As in the two-period model, we assume that investors value collateral services at the beginning of each period, before information about tax revenues and government bond payoffs arrives. Therefore, we condition the expectations operator on the information set available to investors at the end of $t - 1$.

with discount factor $\beta \in (0, 1)$ and a period utility function that satisfies $v'(\cdot) > 0$ and $v''(\cdot) < 0$. The period budget constraint is given by

$$g_t = \tau y_t + h_t \times \left(q_t (B_t - (1 - \delta) B_{t-1}) - \tilde{\delta} B_{t-1} \right),$$

where B_{t-1} denotes legacy debt at the beginning of period t . The budget constraint and the bond payoff depend on the government's credit status h_t . In the autarky case ($h_t = 0$), the government finances the public good g_t with revenues τy_t only. In the repayment case ($h_t = 1$), it can issue new bonds, but has to make debt payments.

Following [Chatterjee and Eyigungor \(2012\)](#), government bonds are modelled as random-maturity bonds. An outstanding bond matures with constant probability δ , where $0 < \delta \leq 1$, whereas it does not mature with probability $1 - \delta$, pays a fixed coupon χ , and is valued – like newly issued bonds – at current market price q_t . The per unit bond payoff k_t depends on the default decision d_t of the issuing government, where $d_t = 1$ if the government defaults and $d_t = 0$ if it repays.

When entering a period with a good credit status $h_{t-1} = 1$, a default immediately changes the government's credit status to $h_t = 0$. By contrast, if the government enters a period in which $h_{t-1} = 0$, with constant probability ϑ , it is given the offer to repay the constant fraction $\omega \in [0, 1]$ of its debt and immediately leave autarky in return. The indicator variable $\xi_t \in \{0, 1\}$ denotes whether such an offer is received ($\xi_t = 1$) or not ($\xi_t = 0$). Following [Hatchondo et al. \(2016\)](#), if the government does not accept an offer ($d_t = 1$), it remains in autarky but its bond position is still reduced by $1 - \omega$. If the government declines an offer, it will be excluded ($h_t = 0$) until the next period, in which, with probability ϑ , it might get a new chance to settle its debt and leave autarky. The law of motion for the credit status can therefore conveniently be written as

$$h_t = \xi_t (1 - d_t) (1 - h_{t-1}) + (1 - d_t) h_{t-1}, \quad (18)$$

with the (realised) bond payoff given by

$$\begin{aligned} k_t = & \mathbf{1}\{h_{t-1} = 1 \wedge d_t = 0\} \left(\tilde{\delta} + (1 - \delta) q_t \right) + \mathbf{1}\{(h_{t-1} = 1 \wedge d_t = 1) \vee (h_{t-1} = 1 \wedge \xi_t = 0)\} q_t \\ & + \mathbf{1}\{h_{t-1} = 0 \wedge \xi_t = 1 \wedge d_t = 1\} \omega q_t + \mathbf{1}\{h_{t-1} = 0 \wedge \xi_t = 1 \wedge d_t = 0\} \omega (\tilde{\delta} + (1 - \delta) q_t). \end{aligned} \quad (19)$$

Here, the auxiliary parameter $\tilde{\delta} \equiv \delta + (1 - \delta)\chi$ summarises the nominal repayment obligation. Bond issuance is defined by $B_t - (1 - \delta) B_{t-1}$, while $\tilde{\delta} B_{t-1}$ is the period t debt service. Following [Bianchi et al. \(2018\)](#), a default leads to utility costs $\phi(\tau y_t) \geq 0$, which are a function of income, and exclusion from financial markets until debt repayment is settled.⁸

The government does not commit to future debt and default policies. As it is common in the literature, we focus on Markov-perfect equilibria, such that government policy in a given period t only depends on the payoff-relevant aggregate state variables, which consist of the aggregate public debt position B_{t-1} , tax revenues y_t , the government's credit status h_{t-1} , and the offer indicator ξ_t by creditors, which is only relevant if the government is in autarky. In [Appendix B](#), we provide a formal definition of the government's policy problem, the model's equilibrium. We also show that the implied prices of a

⁸Public goods provision equals public revenues in periods of default and autarky and, there is no direct output loss in our model that can be related to GDP.

CDS and an alternative risk-free security that provides convenience yield can be computed in a straightforward fashion. Together with the government bond spread, we use these two additional security prices to calibrate the full model to the data.

5 Calibration

Each model period corresponds to one quarter. Our model calibration is based on Italian data from 2001Q1 to 2012Q4. For the data sources, we refer to Table A.1. We solve the model numerically using value function iteration over a discretised state space. The computational algorithm is described in detail in Appendix C.

5.1 Functional forms and parameter choices

Government. The calibration of the fiscal authority follows the sovereign default literature. For the income process, we impose a log-normal AR(1) specification:

$$\log y_t = \rho_y \log y_{t-1} + \sigma_y \varepsilon_{y,t}, \quad \varepsilon_{y,t} \sim N(0, 1). \quad (20)$$

We estimate the income process parameters using Italian GDP data for 1961Q1 to 2012Q4. All data is in real terms, seasonally adjusted, and de-trended using a linear-quadratic trend. The income process is discretised as proposed by Tauchen (1986). For the period utility function of the government, we assume the specification

$$v(g_t) = \frac{(g_t - \underline{g})^{1-\gamma} - 1}{1-\gamma},$$

with $\underline{g}$ denoting a subsistence level of government spending as in Bocola et al. (2019). We set the government's risk aversion parameter to $\gamma = 2$, which is in line with similar models, such as Hatchondo et al. (2016). To calibrate default risk dynamics, we use the same utility cost function as in Bianchi et al. (2018):

$$\phi(y) = \max\{d_0 + d_1 \log(y), 0\}.$$

Following Chatterjee and Eyigungor (2015), we impose a maximum one-quarter-ahead default probability above which investors are not willing to lend. This rules out extreme debt dilution when the economy is close to default, which can arise in models with long-term debt but is at odds with the data.

Bond structure. Characteristics of the sovereign bond market can be informed from the data directly. The maturity parameter δ is chosen such that – without default – debt has an average maturity of five years, which corresponds to our empirical specification. Although the average maturity of Italian bonds slightly exceeds five years during most parts of the sample, using this maturity has several advantages. Namely, the benchmark bonds actually issued by the Italian Treasury have a maturity of five years at issuance and CDS contracted over five years are the most commonly traded credit derivative. We collect coupon rates of all issues of five-year-BTP bonds between 2001 and 2012 from the Italian Department

Parameter	Value	Source/Target
<i>Government, external parameters</i>		
Persistence ρ_y	0.937	Estimate of equation (20)
Volatility σ_y	0.0092	Estimate of equation (20)
Government risk aversion γ	2	Conventional value
Income tax rate τ	1	Normalisation
<i>Financial markets, external parameters</i>		
Investor discount rate r^{rf}	0.0027	Five-year interest rate swap minus inflation swap
Alternative collateral supply b^{alt}	4.34	German bonds outstanding
Coupon parameter χ	0.0115	Average coupon of five-year treasury bonds
Maturity parameter δ	0.05	Five-year average maturity
Recovery rate ω	0.63	Cruces and Trebesch (2013)
Restructuring offer probability ϑ	0.08	Cruces and Trebesch (2013)
Collateral haircut elasticity μ	0.4	Based on Eurosystem collateral framework
Maximum haircut $\bar{\kappa}$	0.7	Based on Eurosystem collateral framework
<i>Calibrated parameters</i>		
Government discount factor β	0.974	Average debt-to-GDP ratio
Default cost parameter d_0	20	Average CDS spread
Default cost parameter d_1	50	Volatility of CDS spread
Minimum consumption $\underline{g}$	0.86	Volatility of debt-to-GDP ratio
Collateral service weight ζ_1	200	Average convenience yield
Collateral service CARA coefficient ζ_2	1.5	Volatility of convenience yield

Table 1: Parameterisation

of the Treasury. The (value-weighted) coupon rate was 4.62, which translates to a quarterly coupon rate of 1.15. The probability of a country in default receiving an offer for debt restructuring ϑ and the corresponding recovery rate ω are in the range identified by [Cruces and Trebesch \(2013\)](#).

Investor preferences. The first key novel ingredient to our model is investors' valuation of collateral services. We impose a CARA functional form,

$$u(\theta_t) = -\frac{\zeta_1}{\zeta_2} \exp\{-\zeta_2(\theta_t + b^{alt})\},$$

where $b^{alt} = 4.34$ refers to alternative sources of collateral, which we proxy by the supply of German government bonds, normalised with Italian GDP. Due to the properties of the CARA utility function, one could also expand the pool of alternative collateral by French or Dutch bonds and recalibrate ζ_1 accordingly without affecting the results. For simplicity, we assume that the German Bund is free of default risk. It follows that the wedge between bond and CDS prices is given by

$$\Lambda_t = (1 - \kappa_{t-1})\zeta_1 \exp\{-\zeta_2((1 - \kappa_t)m_t B_{t-1} + b^{alt})\},$$

which increases in the collateral service weight ζ_1 and decreases in the amount of available collateral $(1 - \kappa_t)m_t B_{t-1}$. Keeping collateral haircuts constant, the size of the convenience yield is then governed by ζ_1 , while ζ_2 determines the elasticity of the convenience yield with respect to available collateral. Since we use maturity-matched interest rate swaps as a risk-free reference rate to compute government bond spreads in the data, we construct our measure of the risk-free rate based on the average five year interest swap minus maturity-matched inflation swaps and obtain a quarterly rate of $r^{rf} = 0.0027$.⁹

Collateral haircut function. The second key ingredient to our calibration is the relationship between sovereign risk and collateral haircuts. In line with our microfoundation of convenience yield from the possibility to use government bonds as collateral on financial markets, note that a bond is typically pledged for a brief period, usually much shorter than the bond maturity. Since haircuts protect market participants from incurring losses on liquidating the collateral in case of a counterparty default, *short-run* sovereign risk matters for collateral haircuts (see also [De Fiore et al., 2023](#)). Therefore, we map default risk into haircuts using a simple parametric form in spirit of [Bindseil \(2014\)](#):

$$\kappa(\lambda, h) = \begin{cases} \min\{\lambda^\mu, \bar{\kappa}\}, & \text{if } h = 1 \\ \bar{\kappa}, & \text{if } h = 0. \end{cases} \quad (21)$$

Here, λ is the one-quarter-ahead default probability, computed over the same horizon as the maximum default probability beyond which investors are unwilling to lend to the government. This functional form implies that collateral haircuts are zero if the bond is default risk-free. The sensitivity parameter is restricted to $\mu \in (0, 1]$, i.e. haircuts increase in the default probability, which is in line with central bank collateral haircuts ([Bindseil and Papadia, 2006](#)) and with margin requirements on the private repo market ([Armakolla et al., 2019](#)).

The haircut elasticity parameter μ is informed using Eurosystem haircuts applicable between October 25, 2008 and December 31, 2010 (see [Nyborg, 2017](#), Table 5.2). Haircuts in the Eurosystem collateral framework are tied to “credit categories”, which are discrete buckets of one-year-ahead default probabilities. We focus on the treatment of government bonds with a residual maturity between five and seven years, consistent with our calibration, in the Eurosystem Credit Assessment Framework (ECAf, available under [this link](#)). Bonds with an annual default probability of up to 0.1%, corresponding to an A- or higher rating, have been subject to a haircut of 3%. Bonds with an one-year-ahead default probability between 0.1% and 0.4%, corresponding to a BBB-rating, received a haircut of 8%. This yields a haircut gap of 5% between BBB-rated bonds and bonds rated A- or higher. To back out the elasticity parameter $\mu = 0.4$, we relate the haircut gap to the difference in (quarterly) default probabilities between those two rating categories, which we approximate by the midpoint of the BBB rating bucket ($\lambda^{BBB} = 0.06\%$) and $\lambda^A \approx 0\%$.¹⁰ In Section 6.3, we demonstrate that our results are robust to using

⁹The EURIBOR itself is an equilibrium interest rate determined by several forces that are beyond the scope of our model, such as investor fundamentals. We use this measure of the risk-free rate because it provides a more accurate measure of investor funding conditions than German or US government bond yields. Since we provide a structural interpretation of the bond price schedule as the solution to an profit maximisation problem of large financial investors that value the collateral services of government bonds, it is natural to use an interest rate that reflects actual financing conditions of intermediaries.

¹⁰We assume that bonds rated A- or higher are approximately default-risk free over the one year horizon, i.e. real world haircuts on these bonds account solely for duration risk, which is not present in our model due to the risk-neutrality assumption of investors. In this case, we have $\kappa(0.06\%) = (0.06/100)^{0.4} = 5.23\%$. Computing the default-risk differential with respect to

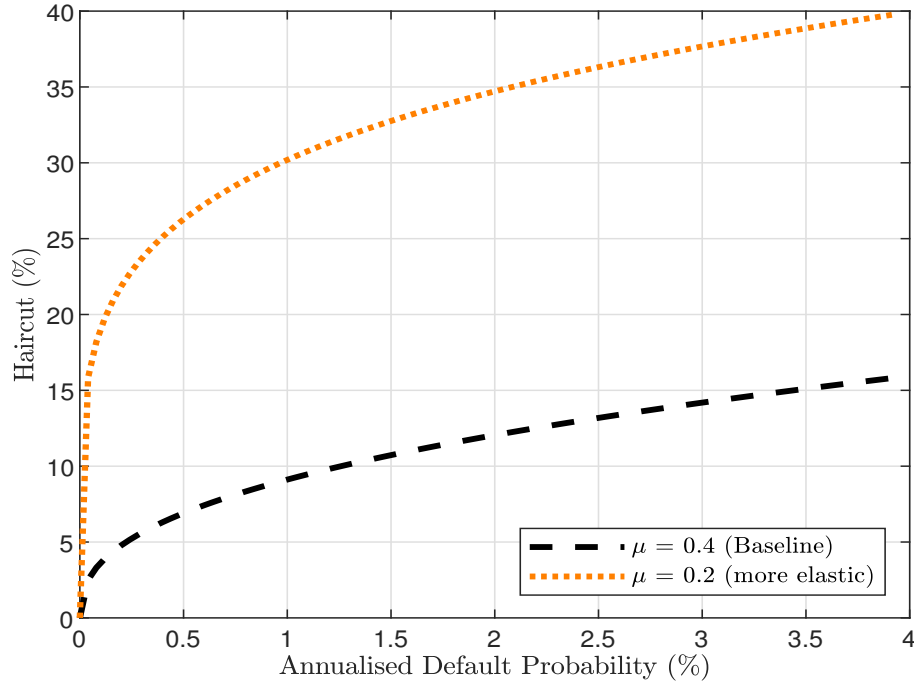


Figure 4: Haircut function

the one-year-ahead default probability in the model, which can be directly mapped into the Eurosystem Credit Assessment Framework.

The baseline haircut schedule is represented by the dotted black line in Figure 4, which shows the applicable collateral haircut for the empirically relevant part of the one-year-ahead default probability, between zero and four percent. Decreasing the elasticity parameter μ implies that haircuts are more responsive to default risk, as demonstrated by the dotted orange line in Figure 4. While under the baseline calibration, a one-year-ahead default probability of 1% implies a haircut of less than 10%, it would be almost 30% for highly elastic haircuts ($\mu = 0.2$). Finally, we apply a ceiling $\bar{\kappa} = 0.7$ to haircuts. This cap is based on the idea that the Eurosystem applied extraordinary haircuts of 35-45% to distressed Cypriot and Greek government bonds (see Nyborg, 2017, Chapter 5.4), while the private repo market for these bonds collapsed (Boissel et al., 2017), which would correspond to a haircut of 100%. Taking the simple mid-point yields $\bar{\kappa} = 70\%$. In the case of default or autarky ($h_t = 0$), the haircut is set to $\bar{\kappa}$ as well. In our simulations, the cap is binding in – at most – 0.7% of all periods in which the government is not in autarky, i.e. the periods based on which we compute the model-implied moments.

Free parameters. The remaining six parameters $(\beta, d_0, d_1, \underline{g}, \zeta_1, \zeta_2)$ are selected to match key moments of financial market variables observed between 2001Q1 and 2012Q4. The government's discount factor β , subsistence consumption $\underline{g}$, and the utility loss parameters d_0 and d_1 are directly linked to borrowing and default incentives. We set $(\beta, d_0, d_1, \underline{g})$ to jointly target the time series average and volatility of government debt issuance and default risk as measured by CDS spreads and the public debt-to-GDP ratio, respectively. These mechanics are well-known in quantitative sovereign default models, see Chatterjee and Eyigungor (2012) or Bocola et al. (2019) for a detailed discussion. Regarding the parameters

the midpoint of A-rating bucket instead of assuming that bonds rated A- or higher are approximately default-risk free would yield a very similar haircut of $\kappa(0.06\% - 0.01\%) = 4.78\%$.

Variable	Full sample mean		Safe episodes		Crisis episodes		Volatility	
	Data	Model	Data	Model	Data	Model	Data	Model
$Q_t B_t / y_t$	4.49	4.47	4.32	4.30	4.81	4.88	0.26	0.38
$c d s_t$	79	75	12	16	201	217	113	174
$c y_t$	32	31	23	22	48	52	31	35
s_t^{rf}	47	44	-11	-7	153	165	115	141
s_t^{DE}	100	92	24	22	246	262	145	194
$basis_t^{DE}$	-21	-18	-9	-6	-45	-45	38	24

Table 2: Model fit – financial market variables. Spreads are annualised and given in basis points, y_t refers to quarterly real GDP. $Q_t B_t$ is the market value of government bonds outstanding. Crisis episodes are all periods with a CDS spread exceeding 40bps. Targeted moments are shown in colour. All model-implied statistics are based on simulations of 500,000 periods, excluding 5,000 burn-in periods. We also exclude all periods in which the government is in financial autarky as well as 40 quarters after reentering financial markets following an exclusion period.

that shape convenience yield, ζ_1 is chosen to match the average convenience yield over the total sample, while the elasticity parameter ζ_2 is chosen to match its volatility. Since we measure convenience yield as the difference between bond spreads over interest rate swaps and CDS spreads, we are implicitly also targeting the level and volatility of the government bond spread. Our calibration is summarised in Table 1.

5.2 Model fit

Table 2 shows the targeted levels and volatilities of CDS spread, convenience yield, and debt-to-GDP ratio as well as the (untargeted) spread over German Bunds and the CDS-bond basis relative to German Bunds, defined in equation (2). Since we have defined convenience yield as the residual from CDS spread and bond spread over investor funding conditions, our model also matches the average bond spread by construction. To examine the interaction between convenience yield and credit risk in greater detail, we also report statistics for *crisis periods*, which we define as all periods with a CDS spread above 40 basis points.¹¹ As the middle columns of Table 2 show, the rise in debt levels, bond spreads and CDS spreads associated with the financial crisis is captured well by our model. Notably, the model also captures an increase in convenience yield, measured vis-à-vis investor funding conditions. At the same time, the spread over the Bund $s_t^{DE} \equiv r_t^{IT} - r_t^{DE}$ is positive throughout the sample and widens dramatically during crisis episodes, consistent with the empirical evidence provided in Section 2. Lastly, we plot kernel density estimations of bond and CDS spreads obtained from our model against their data counterparts in Figure 5. As in Bocola et al. (2019), both spreads show sizable positive skewness and considerable probability mass at negative bond spreads, consistent with the data.

Regarding untargeted moments, the model generates countercyclical debt management, which is typical for developed economies, i.e. the government increases borrowing in response to negative fiscal shocks (see Bocola et al., 2019). It should be noted that such behaviour is not present in standard models of sovereign debt and default (see Arellano, 2008) that are typically calibrated to emerging economies,

¹¹ In the data, this cut-off based on the CDS spread is equivalent to classifying all periods with a positive bond spread as *crisis periods*, as the bond spread was negative from 2001Q1 until 2008Q4 and turned positive thereafter. Using the classification based on the CDS spread allows for a consistent comparison to the model version without convenience yield, where the bond spread is always positive by construction.

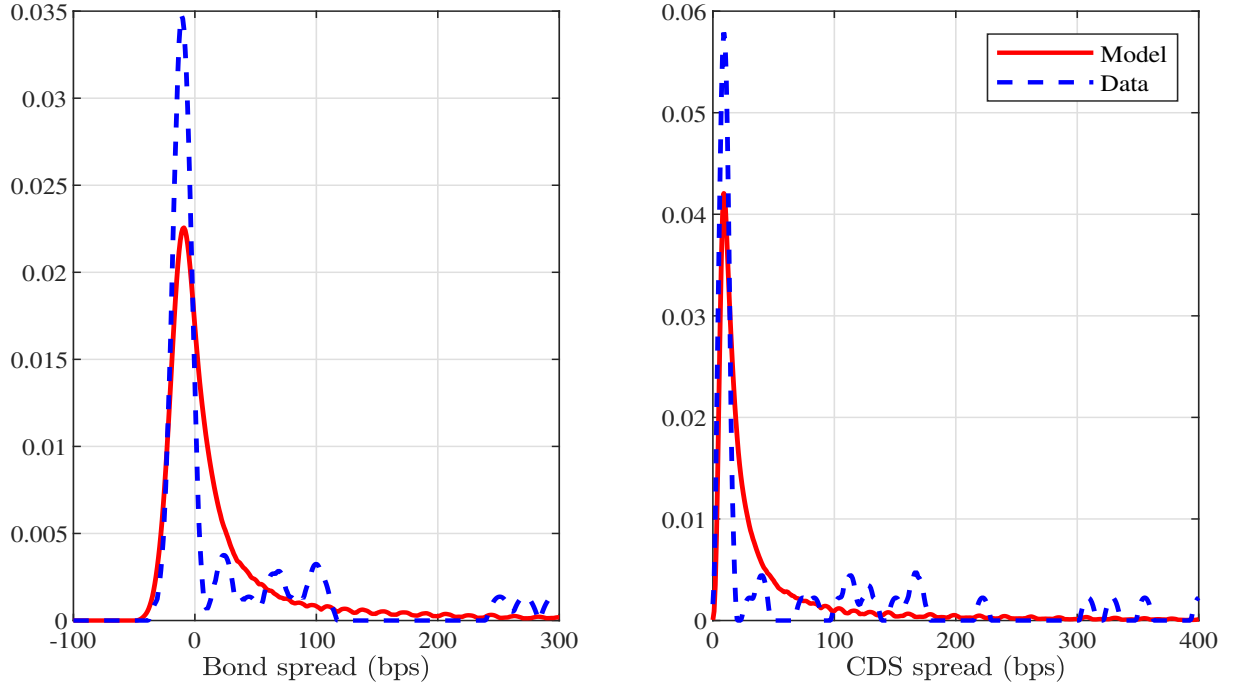


Figure 5: Model fit – distribution of spreads. This figure displays kernel density estimates for bond spreads over the risk-free rate $s_t^{rf} \equiv r_t^{TT} - r_t^{rf}$ (left panel) and CDS spreads (right panel). Spreads are annualised and given in basis points. The moments are shown in colour. The model-implied distribution (solid red line) is based on simulations of 500,000 periods, excluding 5,000 burn-in periods. We also exclude all periods in which the government is in financial autarky as well as 40 quarters after reentering financial markets following an exclusion period. The data moments are from 2001Q1 to 2012Q4 (dashed blue line).

since adverse shocks reduce bond prices due to the increased risk of default, which typically incentivises the government to reduce its debt issuance. However, as in [Bocola et al. \(2019\)](#), the inclusion of a minimum consumption level makes the government less responsive to debt-elastic bond price declines, such that debt issuance and default risk are countercyclical. Consistent with the data, Panel A of Table 3 shows that debt issuance almost doubles in high risk periods. The model also matches the standard deviation of debt issuance reasonably well.

Panel B of Table 3 shows that the countercyclicity of different default risk measures, such as CDS spread and bond spread over the Bund is matched well. Unsurprisingly, such periods are associated with high debt-to-GDP ratios. In addition, the funding disadvantage relative to German Bunds is smaller, i.e. $basis_t^{DE}$ is smaller in absolute terms, if the debt-to-GDP ratio is low. Since the $basis_t^{DE}$ as a measure of the funding disadvantage is always negative, its correlation with debt-to-GDP is negative as well. Likewise, the correlation with CDS spreads as sovereign risk proxy is strongly negative – consistent with the data.

6 Model mechanics: A quantitative exploration

Using our calibrated model, this section quantitatively examines the importance of convenience yield and its components for financial market variables and public debt management. The analysis suggests that the comparative statics exercises in the analytical model from Section 3 carry over to the fully-fledged quantitative model.

Variable	Data	Model
<i>Panel A: debt management</i>		
$\text{ave}(Q_t(B_{t+1} - B_t)/y_t)$	0.029	0.008
$\text{ave}(Q_t(B_{t+1} - B_t)/y_t), \text{ crisis episodes}$	0.045	0.032
$\text{std}(Q_t(B_{t+1} - B_t)/y_t)$	0.049	0.043
$\text{cor}(y_t, Q_t(B_{t+1} - B_t)/y_t)$	-0.22	-0.78
<i>Panel B: cyclical properties</i>		
$\text{cor}(y_t, cds_t)$	-0.68	-0.34
$\text{cor}(y_t, s_t^{DE})$	-0.62	-0.38
$\text{cor}(y_t, basis_t^{DE})$	0.36	0.51
$\text{cor}(Q_t B_{t+1}/y_t, cds_t)$	0.83	0.31
$\text{cor}(Q_t B_{t+1}/y_t, s_t^{DE})$	0.78	0.35
$\text{cor}(Q_t B_{t+1}/y_t, basis_t^{DE})$	-0.52	-0.57
$\text{cor}(cds_t, basis_t^{DE})$	-0.78	-0.78

Table 3: Model fit – additional moments: Spreads are annualised and given in basis points, y_t refers to quarterly real GDP. $Q_t B_t$ is the market value of government bonds outstanding and $Q_t(B_{t+1} - B_t)$ is net debt issuance, at market values. Crisis episodes are all periods with a CDS spread larger than 40 basis points. All model-implied statistics are based on simulations of 500,000 periods, excluding 5,000 burn-in periods. We also exclude all periods in which the government is in financial autarky as well as 40 quarters after reentering financial markets following an exclusion period.

6.1 The role of convenience yield, collateral scarcity, and collateral haircuts

In line with our analysis of the simple model in Section 3, we progressively simplify investors' pricing of collateral services and eventually shut its effects off entirely. In each model variant, we recalibrate the discount factor β and the default cost parameter d_0 to ensure that the average debt-to-GDP ratio and the CDS spread match their data counterparts. We compare the implications of the different model versions in terms of key policy-relevant moments, such as the sovereign default rate and collateral availability. We also examine how convenience and its components affects important financial market variables and public debt management. A pure comparative statics exercise without recalibration is presented in Appendix D.1.

First, we eliminate collateral scarcity by recalibrating the model under the assumption of a constant marginal utility of collateral service. In that case, collateral valuation is given by $\Lambda_t = (1 - \kappa(\lambda_t)) \times \zeta_1$. Since convenience yield still depends negatively on bond supply through the collateral haircut function and default risk, this model version clearly satisfies $\partial \Lambda / \partial b < 0$, corresponding to the dashed black line in Figure 2. We set $\zeta_1 = 0.0007$ to obtain an average convenience yield $cy_t = cds_t - s_t$ of 30bps and recalibrate the model ($d_0 = 21.9141$ and $\beta = 0.964$) to ensure that average CDS and government bond spreads remain consistent with the data moments. Reassuringly, the default rate, average nominal debt-to-GDP ratio (B_t/y_t) and average available collateral (θ_t/y_t) are essentially identical to the baseline model, see the first two columns of Panel A in Table 4.

As panels B to D of Table 4 reveal, an inelastic collateral valuation essentially eliminates all time variation in convenience yield cy_t and the basis over the German Bund ($basis_t^{DE}$). At the same time, the CDS spread exhibits a very similar increase during crisis periods across both model versions. However, as Panel E of Table 4 shows, the absence of a collateral valuation component also implies a much smaller

	(1) Baseline	(2) Constant u' ($\zeta_2 = 0$)	(3) Constant Λ ($\zeta_2 = \mu = 0$)	(4) No CY ($\zeta_1 = 0$)
Marginal convenience yield	$\partial\Lambda/\partial b > 0$	$\partial\Lambda/\partial b < 0$	$\partial\Lambda/\partial b = 0$	-
Discount factor β	0.974	0.964	0.965	0.962
Default cost d_0	20	21.9141	21.5625	24.375
<i>Panel A: Key moments</i>				
Negative spread (%)	49.9	60.4	58.1	0
Default rate (%)	1.33	1.33	1.43	1.40
$\text{ave}(B_t/y_t)$	3.95	3.95	3.93	4.02
$\text{ave}(Q_t B_t/y_t)$	4.47	4.46	4.44	4.50
$\text{ave}(\theta/y_t)$	4.30	4.30	4.40	-
<i>Panel B: Financial markets</i>				
$\text{ave}(cds_t)$	75	75	81	79
$\text{ave}(cy_t)$	31	29	30	0
$\text{ave}(basis_t^{DE})$	-18	0	2	0
<i>Panel C: Safe periods</i>				
$\text{ave}(cds_t)$	16	18	18	18
$\text{ave}(cy_t)$	22	28	28	0
$\text{ave}(basis_t^{DE})$	-6	0	0	0
<i>Panel D: Crisis periods</i>				
$\text{ave}(cds_t)$	217	214	218	214
$\text{ave}(cy_t)$	52	30	33	0
$\text{ave}(basis_t^{DE})$	-45	2	5	0
<i>Panel E: Debt management and cyclical properties</i>				
$\text{ave}(Q_t(B_{t+1} - B_t)/y_t)$	0.008	0.007	0.007	0.007
$\text{ave}(Q_t(B_{t+1} - B_t)/y_t)$, crisis	0.032	0.028	0.028	0.027
$\text{cor}(Q_t(B_{t+1} - B_t)/y_t, y_t)$	-0.778	-0.700	-0.714	-0.714
$\text{cor}(Q_t B_{t+1}/y_t, cds_t)$	0.31	0.09	0.12	0.09
$\text{cor}(Q_t B_{t+1}/y_t, basis_t^{DE})$	-0.57	-0.04	0	0

Table 4: The role of convenience yield – model comparison. This figure compares key model variables across different specifications for convenience yield that we considered in the simple model. Spreads are annualised and given in basis points, y_t refers to quarterly real GDP. $Q_t B_t$ is the market value of government bonds outstanding and $Q_t(B_{t+1} - B_t)$ is net debt issuance, at market values. Crisis episodes are all periods with a CDS spread exceeding 40bps. Targeted moments are shown in colour. All model-implied statistics are based on simulations of 500,000 periods, excluding 5,000 burn-in periods. We also exclude all periods in which the government is in financial autarky as well as 40 quarters after reentering financial markets following an exclusion period.

correlation of CDS spreads and the debt-to-GDP ratio: It declines to 0.09, down from 0.31 in the baseline model and 0.83 in the data. Likewise, the $basis_t^{DE}$ is also almost uncorrelated with the debt-to-GDP ratio, compared to -0.57 in the baseline model and -0.52 in the data, see also Table 3. The reason for this is that investors' collateral valuation does not depend on available collateral and haircuts are essentially zero in most periods, except in the quarters immediately preceding a default event. Therefore, $basis_t^{DE}$

does not become as negative as in the baseline case and in the data during a crisis period, in which the debt-to-GDP ratio is high. It is also worth noting that an elastic collateral valuation permits a stronger countercyclicality of debt issuance, in line with the data. This is directly linked to the positive marginal convenience yield ($\partial\Lambda/\partial b > 0$) in the baseline with elastic collateral valuation, as bond prices *increase* locally if the government borrows to smooth public spending in a downturn.

Lastly, the discount factor has to be reduced from $\beta = 0.974$ to $\beta = 0.964$ in order to match the debt-to-GDP ratio of the baseline calibration. This is consistent with the simple model, where the government optimally chooses a much higher debt level when marginal convenience yield is positive, i.e. with strong collateral scarcity effects, than in the case of a negative marginal convenience yield, i.e. without collateral scarcity effects. In order to render the same debt choice optimal, the government has to be much more impatient.

In the third column, we study the case of an entirely debt-inelastic convenience yield ($\partial\Lambda/\partial b = 0$), which obtains when setting the elasticity parameter in the haircut function (21) to $\mu = 0$. While this model still matches the time series averages of convenience yield, debt-to-GDP ratio and CDS spreads, it misses the time variation of convenience yield. Notably, the discount factor has to be recalibrated only to $\beta = 0.965$, consistent with our theoretical result that a debt-inelastic convenience yield provides larger incentives to issue debt than in the case of a negative marginal convenience yield (see Figure 2). The last column in Table 4 shows that eliminating convenience yield altogether by setting $\zeta_1 = 0$ requires us to reduce the discount rate even further to $\beta = 0.962$.

6.2 Comparative statics: Collateral haircuts, fiscal discipline and welfare

Guided by our analytical decomposition of convenience yield into a collateral scarcity and a collateral haircut component in Section 3, we conduct a comparative statics exercise with respect to the sensitivity of collateral haircuts to default risk, governed by the parameter μ . While the shape of investors' collateral valuation is a deep structural parameter in our model, μ can be interpreted as a policy parameter, at least to the extent that collateral haircuts are applied by central banks or mandated by regulators. This analysis is therefore directly relevant for policymakers if they wish to design a collateral policy that induces disciplinary fiscal policy incentives while at the same time maintaining a high availability of collateral. It turns out that investors' collateral valuation function crucially affects the transmission of collateral policy to sovereign bond market outcomes.

In Table 5, we report the results of increasing the sensitivity of collateral haircuts to default risk for the baseline model and the model with constant collateral valuation. As the left panel shows, reducing μ from 0.4 (its baseline value) to 0.1 increases the default rate from 1.33% p.a. to 1.47%. Since nominal debt outstanding B_t/y_t also increases, this can be interpreted as an erosion of fiscal discipline. By contrast, the right panel shows that the default rate and nominal debt outstanding declines in the model version with inelastic collateral valuation. The results of this exercise are in line with the simple model discussed in Section 3. More sensitive haircuts increase the default frequency in the baseline model with debt-elastic collateral valuation, while they reduce the default frequency in the model with constant collateral valuation.

The last two rows of Table 5 illustrate how changes in the collateral haircut parameter affect welfare of investors and the government. It should be noted the effect sizes cannot be directly compared across

	<i>Baseline model</i>			<i>Constant coll. valuation</i>		
	$\mu = 0.05$	$\mu = 0.2$	$\mu = 0.4$	$\mu = 0.05$	$\mu = 0.2$	$\mu = 0.4$
Default rate (%)	1.47	1.40	1.33	1.23	1.26	1.33
$\text{ave}(B_t/y_t)$	4.04	3.99	3.95	3.93	3.94	3.94
$\text{ave}(\theta_t/y_t)$	4.29	4.30	4.30	4.28	4.30	4.30
Δ Government welfare (%)	-0.70	-0.18	-	+0.40	+0.29	-
Δ Investor welfare (%)	-19.00	-3.25	-	-0.45	+0.10	-

Table 5: Collateral haircut and collateral valuation components – selected moments. This table displays the effects of varying the collateral haircut parameter μ on key variables in different model versions. A low μ corresponds to a highly elastic haircut schedule and its baseline value $\mu = 0.4$ is indicated in color. B_t is the nominal value and θ_t is the haircut-corrected market value of outstanding government bonds. y_t refers to quarterly GDP. All model-implied statistics are based on simulations of 500,000 periods, excluding 5,000 burn-in periods. We also exclude all periods in which the government is in financial autarky as well as 40 quarters after reentering financial markets following an exclusion period. The (annualised) default rate is based on periods where the government has access to financial markets. The welfare changes are computed relative to the baseline case of $\mu = 0.4$. Government welfare is defined in equation (17) and we operationalize investor welfare as expected collateral services $\mathbb{E}_0[u(\theta_t)]$.

model versions, as investors' utility function differs and the government's discount factor takes on different values in each version. Qualitatively, a lower default rate is associated with higher government welfare in either case. Investor welfare depends on the average haircut-corrected market value of government bonds outstanding and also on its variation over time if collateral valuation is debt-elastic. In the baseline model, for example, the government issues more bonds if μ is low, which raises available collateral. However, the average haircut increases as a byproduct, which reduces $\theta_t = (1 - \kappa_t)Q_t B_t$ on average (see also the third row) and makes it more variable over time, as θ_t drops sharply during default events. Hence, investor welfare declines as well when haircut schedules are very sensitive to default risk.

When collateral valuation is kept constant, a highly sensitive haircut schedule ($\mu = 0.05$) raises government welfare, while investor welfare declines. This is in line with the conventional view that more elastic haircuts provide disciplining incentives to fiscal policy, which comes at the cost of lower collateral availability. Varying μ in the baseline model, which provided a much better fit to the data, calls such a clear trade-off into question, at least for large borrowers that react strategically to haircut schedules and investors' valuation of collateral services. Government and investor welfare are highest if haircut schedules are lenient. The endogeneity of collateral valuation and collateral haircuts is key to this result.

Having illustrated the welfare implications of haircut schedules, it has to be stressed that we do not aim to study *optimal* collateral haircut schedules. Our model focuses on the effects of collateral haircuts on debt issuance and default risk of the issuers of collateral. The model does not feature costs that the regulator, e.g. a central bank or a central clearing counterparty (CCP), incurs from accepting risky assets as collateral, which is an essential component of an optimal collateral policy trade-off (Bindseil and Papadia, 2006; Bindseil, 2014; Hall and Reis, 2015). Deriving an appropriate welfare function is not trivial in our setting, as it requires to specify (i) who bears these costs in the end and (ii) how relevant each agent is to the regulator's objective, which ultimately depends on institutional details and also on the residence of investors. We leave these issues for future research.

Taken together, our experiments indicate that the composition of effective convenience yield matters for sovereign risk and the conduct of fiscal policy. Altering the responsiveness of haircut schedules can change default rates by up to 0.1 percentage point. This effect is quantitatively relevant, given that the default rate is less than 1.5% p.a. in our baseline calibration. At the same time, the endogeneity of collateral supply implies that haircut schedules do not visibly affect collateral availability to investors.

6.3 Robustness

This section shows the robustness of our main comparative statics exercise with respect to the elasticity of collateral haircuts. Specifically, we show that the results of the exercise also apply to extensions of our model with respect to the investor pricing kernel and to calibrating the model to other euro area countries. First, we consider the case where haircuts depend on the one-year-ahead instead of the one-quarter-ahead default probability. Second, we augment the model to include trading frictions to show that our results are not driven by market illiquidity as a latent factor. Third, we allow for risk-averse investors, such that bond prices become sensitive to investor consumption, which is affected by default and movements the supply and the price of government bonds. Fourth, we allow for a negative correlation between collateral demand and government revenues. Fifth, we consider the case of countercyclical supply of alternative collateral. In the last two extensions, convenience yield is *ceteris paribus* high (low) in times of crisis, which might alter the governments debt issuance and repayment incentives. Finally, we recalibrate our baseline model to Spain and Portugal.

One-year-ahead default probability. In the baseline model, we have evaluated the haircut function at the one-period-ahead default probability. This corresponds to one quarter in the model and is consistent with the cap on default probabilities (Chatterjee and Eyigungor, 2015). We show that our results are robust to assuming that collateral haircuts depend on the one-year-ahead default probability, which can be computed easily in the model. Notably, this requires a re-parameterisation the haircut parameter to $\mu = 0.5$, using the 0.25% gap between one-year-ahead default probabilities of BBB-rated bonds and bonds rated A- or higher.

Market illiquidity. The literature has identified market illiquidity as a potentially important determinant of government bond prices, in particular in crisis times (Pelizzon et al., 2016).¹² Since illiquidity discounts increase with sovereign risk (see Passadore and Xu, 2022; Chaumont, 2024), they can additionally amplify the widening of government bond spreads in times of high default risk, which in turn affects the sovereigns decision *ex ante*. In an extension, we introduce trading frictions that reflect additional costs investors face when selling or buying a security on secondary markets (Lagos and Rocheteau, 2009). This notion of market illiquidity is conceptually distinct from the collateral services that give rise to convenience yield in our model, see also the discussion in Geromichalos et al. (2023).

In this model version, each period is divided into two sub-periods. In the first sub-period, investors are subject to uninsurable idiosyncratic shocks to the valuation of collateral services. However, we assume that the government bond market is not open at this stage and investors can only trade bonds

¹²It is important to note that government bond markets are usually characterised by exceptionally high liquidity, such that bid-ask spreads are typically very low. Bid-ask spreads are a common measure of market illiquidity, whose interaction with bond spreads is well established (see e.g. He and Milbradt, 2014).

	<i>Baseline</i>			<i>One-year-ahead def. prob.</i>		
	$\mu = 0.05$	$\mu = 0.2$	$\mu = 0.4$	$\mu = 0.05$	$\mu = 0.2$	$\mu = 0.5$
Default rate (%)	1.47	1.39	1.33	1.53	1.48	1.44
$\text{ave}(B_t/y_t)$	4.03	3.98	3.95	4.01	3.96	3.94
$\text{ave}(\theta_t/y_t)$	4.29	4.29	4.29	4.25	4.28	4.27
	<i>Market illiquidity</i>			<i>Risk aversion</i>		
	$\mu = 0.05$	$\mu = 0.2$	$\mu = 0.4$	$\mu = 0.05$	$\mu = 0.2$	$\mu = 0.4$
Default rate (%)	1.47	1.39	1.33	1.55	1.48	1.44
$\text{ave}(B_t/y_t)$	4.03	3.98	3.95	4.13	4.08	4.05
$\text{ave}(\theta_t/y_t)$	4.29	4.29	4.29	4.34	4.35	4.34
	<i>Time-varying altern. coll.</i>			<i>Time-varying coll. valuation</i>		
	$\mu = 0.05$	$\mu = 0.2$	$\mu = 0.4$	$\mu = 0.05$	$\mu = 0.2$	$\mu = 0.4$
Default rate (%)	1.61	1.47	1.44	1.43	1.39	1.35
$\text{ave}(B_t/y_t)$	4.08	4.00	3.98	4.09	4.05	4.02
$\text{ave}(\theta_t/y_t)$	4.30	4.33	4.31	4.39	4.38	4.37
	<i>Spain</i>			<i>Portugal</i>		
	$\mu = 0.05$	$\mu = 0.2$	$\mu = 0.4$	$\mu = 0.05$	$\mu = 0.2$	$\mu = 0.4$
Default rate (%)	1.85	1.81	1.81	0.94	0.94	0.89
$\text{ave}(B_t/y_t)$	1.70	1.69	1.69	2.69	2.70	2.69
$\text{ave}(\theta_t/y_t)$	1.76	1.80	1.80	2.94	2.96	2.95

Table 6: The effect of collateral haircuts in different model extensions. This table displays the effects of varying the collateral haircut parameter μ on key variables in various model extensions and robustness tests. A low μ corresponds to a highly elastic haircut schedule and its baseline value $\mu = 0.4$ is indicated in color. B_t is the nominal value and θ_t is the haircut-corrected market value of outstanding government bonds. y_t refers to quarterly GDP. All model-implied statistics are based on simulations of 500,000 periods, excluding 5,000 burn-in periods. We also exclude all periods in which the government is in financial autarky as well as 40 quarters after reentering financial markets following an exclusion period. The (annualised) default rate is based on periods where the government has access to financial markets.

with dealers on a decentralised over-the-counter market.¹³ At the beginning of the second sub-period, tax revenues realise, the government decides to default or how much to borrow and the competitive and centralised government bond market is open.

We assume that investors are subject to i.i.d. preference shocks, which implies that there are gains from trade in the first sub-period. With a probability of 1/2, investors are either a high-collateral-valuation or a low-collateral-valuation type. Investors cannot directly trade bonds among each other or with the government in the first sub-period, but can instead adjust their bond holdings by trading with dealers as in [Lagos and Rocheteau \(2009\)](#). Restricting their ability to trade with each other affects the pricing of government bonds in the second sub-period and, as a by-product, gives rise to endogenous bid-ask spreads. The maximisation problem, presented in detail in [Appendix D.2](#), yields a bond pricing

¹³Dividing a period into two sub-periods that represent a decentralised and a centralised market, respectively, is in the spirit of [Lagos and Wright \(2005\)](#).

condition with an extended expression for convenience yield:

$$\Lambda_{t+1} \equiv \underbrace{(1 - \kappa_{t+1})}_{1-\text{haircut}} \times \underbrace{\mathbb{E}_i \left[u'_i(\theta_{t+1}) + \frac{1}{2} \times \left\{ u'_i(\tilde{\theta}_{i,t+1}) - u'_i(\theta_{t+1}) \right\} \right]}_{\text{collateral valuation}}. \quad (22)$$

The term in curly brackets reflects trading frictions when dealers have bargaining power vis-à-vis investors. Specifically, the new term $u'_i(\tilde{\theta}_{i,t+1}) - u'_i(\theta_{t+1})$ measures the net gains from trading, with $\tilde{\theta}_{i,t+1}$ denoting the collateral value of the bond position held by an investor of type i after trading on the decentralised market, see Appendix D.2 for details. Without preference heterogeneity, equation (22) collapses to the expression in the baseline case (16).

Since market illiquidity is tied to convenience yield in our model, the CDS price does not contain illiquidity discounts. To still disentangle convenience yield and market illiquidity, we instead exploit the analytical tractability of the decentralised/centralised market framework. As outlined in Appendix D.2, we can express the bid-ask spread on government bonds in closed form. To quantify the effects of market illiquidity, we extend the collateral valuation function by two parameters that enable us to match both the level and volatility of bid-ask spreads.

Investor risk aversion. Our baseline model assumes that investors value pecuniary payoffs in a risk-neutral fashion according to their expected value. We show that our key results are robust to relaxing this assumption and considering risk-averse investors. While investor consumption already depends on the government's net debt issuance and on default risk in the baseline model, this did not enter the bond pricing condition due to the quasi-linear preferences. With risk aversion, bond price schedules are affected by investors' (marginal) utility under the respective debt choices. The conceptual details and additional information on the substantially more complicated equilibrium computation are relegated to Appendix D.3.

Time-varying collateral valuation. In the baseline model, we have implicitly assumed that adverse shocks to fiscal conditions are uncorrelated with investors' collateral valuation, which is conceptually reasonable under the assumption that sovereign risk is driven exclusively by fiscal shocks that are orthogonal to investors' financial conditions. However, since the European sovereign debt crisis was associated with stress on private financial markets, this simplification can be called into question. For example, investors might need more collateral during a sovereign debt crisis to borrow on the secured interbank market segment or to participate in lender-of-last-resort policies by central banks (Drechsler et al., 2016).¹⁴ This increases the valuation of collateral services from government bond holdings during crisis times. We show that our main comparative statics exercise does not depend critically on *time-varying collateral valuation*.¹⁵ Ceteris paribus, this raises the bond price in recessions, but induces the government to issue more debt ex ante. Formally, this corresponds to making collateral demand depen-

¹⁴The positive relationship between collateral valuation and private sector financial stress is discussed in Vayanos (2004). See also Corradin et al. (2017) and the references therein. A similar feature of investors' bond pricing condition also arises in the framework of Brunnermeier et al. (2024): If recessions are associated with higher idiosyncratic risk, the valuation of government bonds' safety attributes increases during downturns.

¹⁵A similar cyclicity of convenience yield can also be microfounded by countercyclical idiosyncratic income risk at the household level (Brunnermeier et al., 2024)

dent on the exogenous revenue state, $u(\theta_t, y_t)$, such that $u'(\theta_t, y_t^{low}) > u'(\theta_t, y_t^{high})$. Appendix D.4 shows the functional form specifications, the parameterisation and demonstrates that this model extension can enhance the model's ability to replicate the empirical relationship between bond supply and convenience yield.

Time-varying alternative collateral. In the baseline model, we have calibrated the supply of alternative collateral b^{alt} to the full-sample average of (real) German debt, relative to Italian GDP which is our proxy for government income. In practice, fiscal policy in Germany was counter-cyclical, such that its debt-to-GDP ratio increased during the Great Financial Crisis and the euro area debt crisis. As business cycles in Italy and Germany exhibit substantial comovement, the supply of alternative collateral is particularly large and the collateral value of Italian government bonds is (ceteris paribus) particularly low if public revenues are low. Hence, it comes natural to examine whether this has implications for our main comparative statics exercises with respect to collateral haircut schedules. Different to the extension with time-varying collateral valuation, the bond price (ceteris paribus) declines in recessions, which induces government to issue less debt ex ante. In this extension, we demonstrate that our results are robust to assuming that German debt alternative collateral is supplied counter-cyclically. We provide additional details on the parameterisation and model fit in Appendix D.5.

Other euro area countries. Our main quantitative results are based on a calibration to Italy. However, empirical evidence in Appendix A shows that key empirical features of convenience yield, sovereign risk and public debt management are also present in other euro area member states. Motivated by this observation, we show that our key comparative statics exercise concerning the elasticity of haircut schedules is robust to calibrating the model to Spain or Portugal instead. In line with the empirical analysis in Bocola et al. (2019), we focus on these countries, because they are the second and third largest country that experienced large default risk in the European debt crisis. For the details on the recalibrated models, we refer to Appendix D.6. While both countries exhibit different CDS spread levels and debt-to-GDP ratios than our baseline calibration to Italy, the main interactions between collateral valuation and collateral haircut effects are also present in both recalibrations.

Results. Table 6 summarises our main comparative statics experiment with respect to haircut elasticities for the five model modifications and extensions as well as the calibrations for Spain and Portugal. The case with market illiquidity and one-year-ahead default probabilities entering the haircut function only show very small differences compared to the baseline model. Likewise, the model versions with time-varying convenience yield, time-varying supply of alternative collateral, and with investor risk aversion deliver the same comparative statics as our baseline model, although the differences to the baseline model are slightly larger in these cases. The same holds true for the model calibrations based on data for Spain and Portugal.

7 Conclusion

In this paper, we studied how convenience yield, sovereign default risk, and the supply of government bonds interact from a macroeconomic perspective. First, we provided empirical evidence on this inter-

action for euro area government bonds. Specifically, we showed that convenience yield of these bonds (i) increases in their relative scarcity and (ii) declines with default risk, which is consistent with default-risk sensitive haircuts applied to collateral on financial markets. Second, we showed for an analytically tractable two-period model that convenience yield interacts non-trivially with sovereign risk and the supply of government bonds. The tractable model demonstrates that the debt-elasticity of convenience yield is a crucial driver of sovereign bond market outcomes and provides an intuitive decomposition into the debt-elasticity of collateral valuation and collateral haircuts. Third, we study an infinite-horizon version of the model to investigate the quantitative relevance of our analytical results. Despite its parsimonious structure, our model can generate the main observed properties of sovereign debt, credit risk, and sovereign bond spreads. Contrary to conventional wisdom, we found that more debt-elastic collateral haircut schedules fail to provide fiscal discipline in our baseline model, as strong collateral scarcity effects increase bond prices during a downturn. This property incentivises the government to issue more debt ex ante and increases the default rate ex post. Our analysis thus suggests that the design of collateral haircut schedules benefits from taking investors' collateral valuation and endogenous debt supply responses by the government into account.

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A Data appendix

This section contains the data sources and provides additional descriptive statistics.

Income process. The GDP data for Italy is obtained from the St Louis Fed database, going from 1961Q1 to 2012Q4. The data is in real terms and seasonally adjusted. The data is logged and de-trended using deviations from a linear-quadratic trend. We subsequently impose an AR(1) structure and obtain $(\rho_y, \sigma_y) = (0.937, 0.0092)$.

Series	Source
Turnover, five-year BTP	Bank of Italy
Net issuance, long-term	Bank of Italy
Turnover, total debt	Bank of Italy
Debt service, long-term	Department of the Treasury monthly reports
Bid-ask spreads, five-year BTP	Department of the Treasury quarterly bulletin
EURIBOR swap, five-year	Bloomberg
Inflation swap, five-year	Bloomberg
CDS spreads, five-year	Bloomberg
Government bond spreads, five-year	Bloomberg
Total debt outstanding	Eurostat
Euro area CPI	St Louis Fed
Quarterly GDP	St Louis Fed

Table A.1: Data sources and ticker

Investors. As a proxy of the (real) discount rate of investors, we use five year interest swaps based on the three-month EURIBOR (2001Q1 to 2012Q4) and five year inflation swaps, both obtained from Bloomberg. The real rate is simply computed by subtracting quarterly (expected) inflation from the interest rate swap, which yields a full sample average of 106 basis points (annualised) and translates into $r_t^{rf} = 0.0027$.

Debt. Debt service is taken from monthly reports of the Italian Department of the Treasury. To focus on debt with appropriate maturities, we focus on redemption and coupon payments of bonds with maturities of at least one year. The inclusion of short-term liabilities (Treasury Bills, referred to as BOT) would introduce a relatively large amount of outstanding debt that is potentially rolled over multiple times each period. Coupon data are obtained from annual reports on debt issuance also published by the Treasury.

Bond Spreads. We obtain government bond yields, CDS spreads and interest rate swaps from Bloomberg, while debt-to-GDP ratios are downloaded from Eurostat. Appendix A.1 shows the time series of all spreads relevant for our quantitative analysis. Both the CDS spread (red) the bond spread over German Bunds (dashed blue) are positive throughout the sample. While closely moving together until 2010, the government bond spread over the Bund experiences a much larger increase once spreads exceed 200bps. A similar pattern can be observed in Spain, Portugal, Ireland, and Greece. This drastic widening of the

$basis_t^{DE,j}$ reflects the large relative convenience yield on German Bunds, who maintained their AAA-rating throughout the sample. By contrast, the bond spread over investor funding conditions (measured by the IRS, dotted black line) is negative until the beginning of 2009, which also marks the beginning of the crisis episode used in the calibration. In all five countries, the bond spread over the risk-free rate s_t^{rf} turns positive and is close to the CDS spread once default risk becomes pronounced.

Complementary to Figure 1 in the main text, we display the relationship between convenience yield and default risk for Spain, Portugal, Ireland, and Greece in Appendix A.2. These countries have all been characterized by large sovereign risk and sharply rising public debt-to-GDP ratios after the Great Financial Crisis of 2008. In all cases, there is a strong negative relationship between CDS spreads and debt-to-GDP ratio with the CDS-bond basis over the Bund, respectively.

In Table A.2, we demonstrate that the key data moments used in the quantitative analysis are very similar across all peripheral euro area countries. For all countries, we observe substantially larger debt-to-GDP ratios, CDS spread and a widening in the CDS-bond basis vis-à-vis Germany, in the high risk periods. Importantly, the classification into low and high risk periods is differs at most by one quarter between all five countries, as the government bond spread over the risk-free rate turned positive in 2008 across the euro area periphery. While the basis over the Bund is very similar in the safe period, somewhere between -4bps and -9bps, it widens to around -50bps in Spain and Italy. The numbers are somewhat larger for Greece, which is not surprising given that Greece went into debt restructuring. Consequently, the convenience yield on Greek government bonds declined to zero during the crisis episode, while it exhibits a similar increase during the crisis period for the other four countries.

Moment	Italy	Spain	Portugal	Ireland	Greece
$\text{ave}(Q_t B_t / y_t)$	4.49	2.00	3.12	1.94	4.47
$\text{ave}(Q_t B_t / y_t \text{safe}_t)$	4.32	1.74	2.60	1.17	3.76
$\text{ave}(Q_t B_t / y_t \text{crisis}_t)$	4.81	2.46	4.06	3.34	5.76
$\text{ave}(cds_t)$	79	78	58	308	100
$\text{ave}(cds_t \text{safe}_t)$	15	8	11	21	16
$\text{ave}(cds_t \text{crisis}_t)$	206	206	188	375	389
$\text{ave}(cy_t)$	32	33	33	73	18
$\text{ave}(cy_t \text{safe}_t)$	23	28	26	54	22
$\text{ave}(cy_t \text{crisis}_t)$	48	43	52	78	0
$\text{ave}(basis_t^{DE})$	-21	-20	-11	-13	-26
$\text{ave}(basis_t^{DE} \text{safe}_t)$	-9	-4	-4	-5	-9
$\text{ave}(basis_t^{DE} \text{crisis}_t)$	-46	-50	-29	-15	-82

Table A.2: Data moments for euro area periphery countries. The basis vis-à-vis the Bund ($basis_t^{DE}$), the convenience yield cy_t , and the CDS spread cds_t are measured in basis points. y_t refers to quarterly GDP. $Q_t B_t$ is the market value of government bonds outstanding. Crisis episodes are all periods with a CDS spread exceeding 40bps. For Greece and Portugal, we truncate the sample in 2011Q1 after Portugal received a bailout by the European Financial Stability Facility. For Ireland, CDS data are only available after 2007, such that we refrain from truncating the sample early.

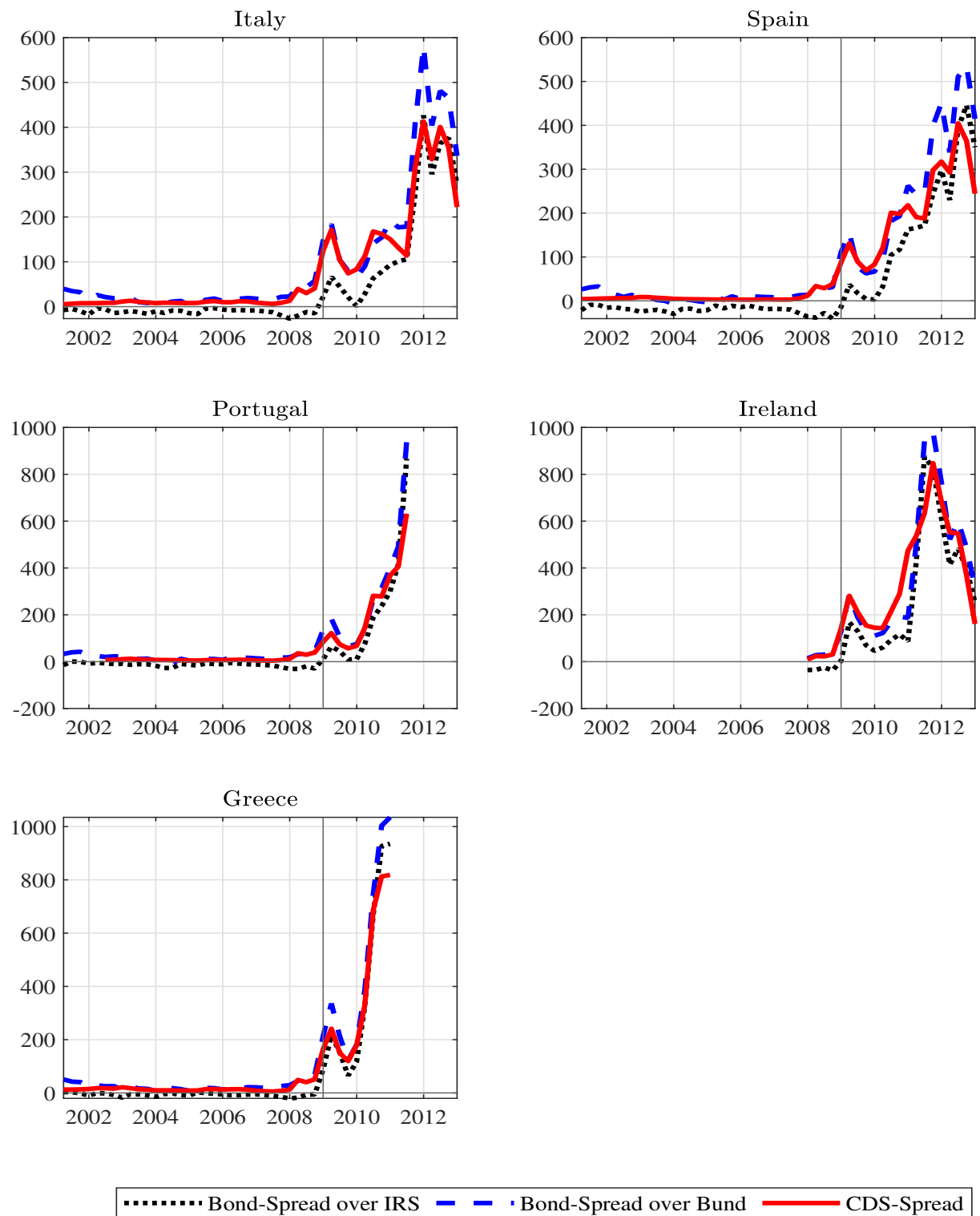


Figure A.1: Government bond and CDS spreads in basis points over time.

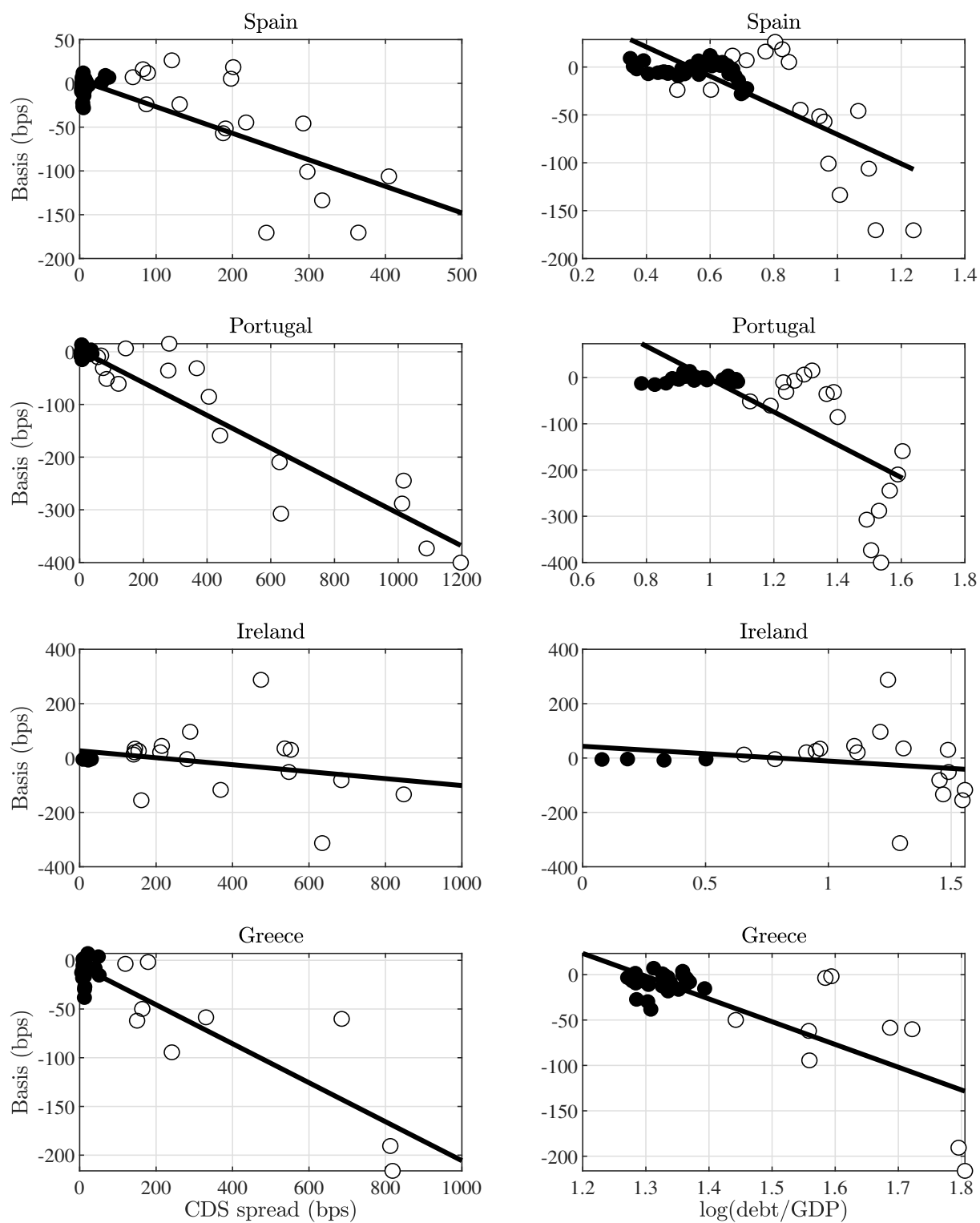


Figure A.2: Relationship between convenience yield, bond supply and default risk in other euro area countries. The left panel plots the basis vis-à-vis the Bund against the CDS spread, both measured in basis points. The right panel plots the basis against $\log(\text{debt}/\text{GDP})$, which is computed based on quarterly GDP. The pre-crisis sub-sample is indicated by black dots, the post-crisis sub-sample by white dots.

B Model Appendix: Policy problem and equilibrium definition

In this section, we formally define the government's policy problem and the model's equilibrium for the baseline model introduced in Section 4. We present the policy problem recursively, with the next period's variables indicated by a prime. In the repayment case, assuming bond market clearing ($b = B$), the government faces the following bond pricing schedule for an arbitrary debt choice B' , and current income y :

$$q(B', y) = \frac{1}{1 + r^f} m^r(B', y) (1 + \Lambda^r(B', y)) . \quad (\text{B.1})$$

The bond price schedule consists of two terms. The first one, given by the function

$$m^r(B', y) = \mathbb{E}_{y'|y} \left[(1 - \mathcal{D}(B', y')) \left(\tilde{\delta} + (1 - \delta) \mathcal{Q}^r(B', y') \right) + \mathcal{D}(B', y') \mathcal{Q}^d(B', y') \right] ,$$

captures the expected pecuniary value of the bond in the next period, where $\mathcal{D}(\cdot)$ denotes the default policy function. Bond prices exceed the expected pecuniary value, which is reflected in the convenience yield

$$\Lambda^r(B', y) = (1 - \kappa(\lambda(B', y), 1)) \times u'(\Theta^r(B', y)) .$$

Here, the term

$$\Theta^r(B', y) = (1 - \kappa(\lambda(B', y), 1)) \times m^r(B', y) \times B' ,$$

is the effective collateral service, which depends on the government's debt issuance B' . The collateral haircut $\kappa(\lambda(B', y), 1)$ only depends on the default probability and the government's credit status. The default probability $\lambda(B', y)$ follows from the one-period-ahead default policy

$$\lambda(B', y) = \mathbb{E}_{y'|y} [\mathcal{D}(B', y')] .$$

The functions $\mathcal{Q}^r(\cdot)$ and $\mathcal{Q}^d(\cdot)$ determine the equilibrium bond prices in the next period for the repayment and default case. The equilibrium bond price in the repayment case is obtained by evaluating the pricing schedule at the debt policy function $\mathcal{B}(\cdot)$

$$\mathcal{Q}^r(B, y) = q(\mathcal{B}(B, y), y) . \quad (\text{B.2})$$

In the default (and autarky) case, debt service is suspended and the debt position is rolled over, such that the associated equilibrium bond price is given as

$$\mathcal{Q}^d(B, y) = \frac{1}{1 + r^f} \left(m^d(B, y) (1 + \Lambda^d(B, y)) \right) , \quad (\text{B.3})$$

where the pecuniary benefits now are given by

$$m^d(B', y) = \vartheta \omega m^r(\omega B', y) + (1 - \vartheta) \mathbb{E}_{y'|y} [\mathcal{Q}^d(B', y')] ,$$

and the non-pecuniary ones by

$$\Lambda^d(B', y) = (1 - \kappa(1, 0)) \times u'(\Theta^d(B', y)) ,$$

with collateral services

$$\Theta^d(B', y) = (1 - \kappa(1, 0)) \times m^d(B', y) \times B' .$$

The government's problem can be written recursively. When not in autarky, the government decision problem is given by

$$\mathcal{F}(B, y) = \max_{d \in \{0, 1\}} (1 - d) \mathcal{F}^r(B, y) + d \mathcal{F}^d(B, y) , \quad (\text{B.4})$$

where the value of repayment $\mathcal{F}^r(B, y)$ satisfies the Bellman equation

$$\mathcal{F}^r(B, y) = \max_{B'} v \left(\tau y + q(B', y) (B' - (1 - \delta) B) - \tilde{\delta} B \right) + \beta \mathbb{E}_{y'|y} [\mathcal{F}(B', y')] , \quad (\text{B.5})$$

and the value of default (and autarky) $\mathcal{F}^d(B, y)$ satisfies

$$\mathcal{F}^d(B, y) = v(\tau y) - \phi(\tau y) + \beta \mathbb{E}_{y'|y} \left[\vartheta \mathcal{F}(\omega B, y') + (1 - \vartheta) \mathcal{F}^d(B, y') \right] . \quad (\text{B.6})$$

The equilibrium for the model is defined as follows.

Equilibrium. A Markov-perfect equilibrium consists of value, policy, and bond price functions $\{\mathcal{F}(\cdot), \mathcal{F}^r(\cdot), \mathcal{F}^d(\cdot), \mathcal{B}(\cdot), \mathcal{D}(\cdot), q(\cdot), \mathcal{Q}^r(\cdot), \mathcal{Q}^d(\cdot)\}$, such that

- (i) $\mathcal{F}(B, y)$, $\mathcal{F}^r(B, y)$, and $\mathcal{F}^d(B, y)$ satisfy equations (B.4) to (B.6),
- (ii) $\mathcal{B}(B, y)$ and $\mathcal{D}(B, y)$ solve equations (B.4) and (B.5),
- (iii) $q(B, y)$, $\mathcal{Q}^r(B, y)$ and $\mathcal{Q}^d(B, y)$ satisfy equations (B.1) to (B.3).

Having defined the model equilibrium, it is possible to price a derivative security that will serve as the model equivalent of a CDS. A security reflecting the payoff structure of a CDS can be mapped into our model by removing convenience yield from the pricing condition (B.1):

$$q^{CDS}(B', y) = \frac{1}{1 + r^r f} \mathbb{E}_{y'|y} \left[\begin{aligned} & (1 - \mathcal{D}(B', y')) \left(\tilde{\delta} + (1 - \delta) \mathcal{Q}^{CDS, r}(B', y') \right) \\ & + \mathcal{D}(B', y') \mathcal{Q}^{CDS, d}(B', y') \end{aligned} \right] . \quad (\text{B.7})$$

To obtain the equilibrium CDS price in the repayment and default cases, the CDS pricing condition (B.7) is evaluated at the equilibrium debt policy:

$$\mathcal{Q}^{CDS, r}(B, y) = q^{CDS}(\mathcal{B}(B, y), y) , \quad (\text{B.8})$$

$$\mathcal{Q}^{CDS, d}(B, y) = \vartheta \omega \mathcal{Q}^{CDS, r}(\omega B, y) + \frac{1}{1 + r^r f} \mathbb{E}_{y'|y} \left[(1 - \vartheta) \mathcal{Q}^{CDS, d}(B, y') \right] . \quad (\text{B.9})$$

The crucial difference to the bond pricing expressions represented by equations (B.2) and (B.3) is the absence of collateral service: The government bond will trade at a higher price than the CDS price whenever collateral service is positive. Similarly, we can use the investor pricing condition to compute the model implied price of a German Bund with the same maturity structure, but without default risk:

$$q^{DE}(B', y) = \frac{1}{1 + r^{rf}} \mathbb{E}_{y'|y} \left[\begin{array}{l} (1 - \mathcal{D}(B', y')) \left(\tilde{\delta} + (1 - \delta) \mathcal{Q}^{DE,r}(B', y') \right) \\ + \mathcal{D}(B', y') \left(\tilde{\delta} + (1 - \delta) \mathcal{Q}^{DE,d}(B', y') \right) \end{array} \right]. \quad (\text{B.10})$$

Note that we the credit status of the (Italian) government and the riskiness of their bonds are still relevant for the pricing of the Bund, since they determine the supply of collateral and investors' collateral valuation. In equilibrium, we obtain

$$\mathcal{Q}^{DE,r}(B, y) = q^{DE}(\mathcal{B}(B, y), y) \quad (\text{B.11})$$

$$\mathcal{Q}^{DE,d}(B, y) = \vartheta \mathcal{Q}^{DE,r}(\omega B, y) + \frac{1}{1 + r^{rf}} \mathbb{E}_{y'|y} \left[(1 - \vartheta) \mathcal{Q}^{DE,d}(B, y') \right]. \quad (\text{B.12})$$

Note that the Bund's nominal payoff is not reduced by the factor ω in (B.12) for the case of re-entry. However, the reduction of outstanding (Italian) bonds is still reflected in (B.10) and (B.12) due to its effects on collateral availability.

C Solution algorithm

This section presents details on the numerical solution algorithm used to solve the government problem.

Taste shocks. We solve the model numerically using value function iteration on a discrete state space without interpolation. The debt grid is denoted by $\mathfrak{B}$. The expected continuation value is defined as $F(B', y) \equiv \mathbb{E}_{y'|y} \mathcal{F}(B', y')$. As already pointed out by [Chatterjee and Eyigungor \(2012\)](#), convergence problems typically arise in this class of models. We follow [Gordon \(2019\)](#) by introducing Gumbel-distributed taste shocks z to randomise over debt and default choices:

$$\mathcal{F}^r(B, y', z) = \max_{B'} \left\{ v \left(\tau y + q(B', y) (B' - (1 - \delta)B) - \tilde{\delta}B \right) + \tilde{\beta} F(B', y) + \sigma_z z^{B'} \right\}.$$

Throughout the algorithm we standardise the value function prior to computing choice probabilities and expected choices. The standardisation is immediately cancelled out when choice probabilities are computed. For the expected value of the maximum choice, consider the well-known expression

$$\mathbb{E} \left(\max_j V(j) + \sigma \varepsilon_j \right) = \sigma \log \left(\sum_j \exp \left\{ \frac{V(j)}{\sigma} \right\} \right). \quad (\text{C.1})$$

where j denotes the choice in the next period. Dependence on current persistent states is omitted. Expanding both sides by an arbitrary constant $\bar{V}$ yields

$$\bar{V} + \mathbb{E} \left(\max_j V(j) - \bar{V} + \sigma \varepsilon_j \right) = \sigma \log \left(\sum_j \exp \left\{ \frac{V(j) - \bar{V}}{\sigma} \right\} \right). \quad (\text{C.2})$$

The right-hand side of (C.2) can be rearranged as

$$\begin{aligned} \bar{V} + \sigma \log \left(\sum_j \exp \left\{ \frac{V(j) - \bar{V}}{\sigma} \right\} \right) &= \bar{V} + \sigma \log \left(\exp \left\{ -\frac{\bar{V}}{\sigma} \right\} \sum_j \exp \left\{ \frac{V(j)}{\sigma} \right\} \right) \\ &= \bar{V} + \sigma \log \left(\exp \left\{ -\frac{\bar{V}}{\sigma} \right\} \right) + \sigma \log \left(\sum_j \exp \left\{ \frac{V(j)}{\sigma} \right\} \right) \\ &= \sigma \log \left(\sum_j \exp \left\{ \frac{V(j)}{\sigma} \right\} \right). \end{aligned}$$

Plugging back into (C.2) gives

$$\mathbb{E} \left(\max_j V(j) + \sigma \varepsilon_j \right) = \bar{V} + \sigma \log \left(\sum_j \exp \left\{ \frac{V(j) - \bar{V}}{\sigma} \right\} \right). \quad (\text{C.3})$$

When $\bar{V} = \max_j V(j)$ is chosen, each term in the exponent on the left-hand side is negative and the problem remains numerically tractable. To speed up computation, we use a version of the divide-and-conquer algorithm proposed by [Gordon and Qiu \(2018\)](#) and [Gordon \(2019\)](#) that truncates numerically irrelevant choices based on policy function monotonicity. The improved convergence properties compared to a brute-force algorithm are particularly relevant for the large state space required by our model.

Since there are some (numerically small) non-monotonicities at some points in the state space, we use a guess-and-verify approach. We compute the equilibrium using the divide-and-conquer approach until convergence. After the final step, we use that solution to compute another iteration. We consider the solution obtained using the divide-and-conquer approach appropriate if the equilibrium objects remain reasonably near to the solution from the previous iteration.

Algorithm. To start off the algorithm, we use the final period of a finite-horizon model to determine default states for iteration 0. Given the results of iteration ι (or initialisation), each iteration $\iota + 1$ proceeds as follows:

1. For all states (B, y') , evaluate debt choices B' according to

$$\left\{ \bar{B}, 0, \frac{1}{2}\bar{B}, \frac{3}{4}\bar{B}, \frac{1}{4}\bar{B}, \frac{7}{8}\bar{B}, \frac{5}{8}\bar{B}, \dots \right\}.$$

- (a) Determine the value of default:

$$\mathcal{F}_{\iota+1}^d(B, y') = v(\tau y' - \phi(\tau y')) - \phi(\tau y') + F_t^d(B, y').$$

- (b) Evaluate every numerically relevant debt choice in $\mathfrak{B}$, given current debt over the interval $[B'_-, B'_+]$ (elements of the grid $\mathfrak{B}$) with

$$B'_- = \min \left(B' | P(\mathcal{B}(B_-)) > \varepsilon \right), \quad \text{and} \quad B'_+ = \max \left(B' | P(\mathcal{B}(B_+)) > \varepsilon \right),$$

where B_+ and B_- respectively denote the next largest and smallest current debt stock that has already been evaluated. At the upper bound, the choices are not bounded, while for $B_- = 0$, they are only bounded from above. The value of choosing $B^j \in \mathfrak{B} | B_- < B^j < B_+$ is given by

$$\mathcal{F}_{\iota+1}^r(B^j | B, y') = v \left(\tau y' + \mathcal{Q}_t^r(B^j, y') (B^j - (1 - \delta)B) - \tilde{\delta}B \right) + \tilde{\beta} F_t(B^j, y').$$

- (c) Compute debt choice and default probabilities using the Type I extreme value distribution:

$$\Pr(d = 1 | B, y') = \frac{\exp(\mathcal{F}_{\iota+1}^d(B, y')/\sigma_z)}{\exp(\mathcal{F}_{\iota+1}^d(B, y')/\sigma_z) + \sum_j \exp(\mathcal{F}_{\iota+1}^r(B^j, y')/\sigma_z)},$$

$$\Pr(B' = B^j | B, y') = \frac{\exp(\mathcal{F}_{\iota+1}^r(B^j, y')/\sigma_z)}{\exp(\mathcal{F}_{\iota+1}^d(B, y')/\sigma_z) + \sum_j \exp(\mathcal{F}_{\iota+1}^r(B^j, y')/\sigma_z)}.$$

- (d) Determine the default policy $\mathcal{D}$ and expected payoffs k^r and k^d w.r.t. z using the probabilities computed in (c):

$$\mathcal{D}_{\iota+1}(B, y') = \Pr(d = 1 | B, y'),$$

$$k_{\iota+1}^r(B, y') = \sum_j \Pr(B^j | B, y') \left[\tilde{\delta} + (1 - \delta) \mathcal{Q}_t^r(B^j, y') \right],$$

$$k_{\iota+1}^d(B, y') = \vartheta \omega \left\{ (1 - \mathcal{D}_t(\omega B, P(\omega B, y'), y')) \times \left(\tilde{\delta} + (1 - \delta) \mathcal{Q}_t^r(\omega B, y') \right) + \mathcal{D}_t(\omega B, y') \mathcal{Q}_t^d(\omega B, y') \right\} + (1 - \vartheta) \mathcal{Q}_t^d(B, y'),$$

where the bond prices under reentry $\mathcal{Q}_i^r(\omega B, y')$ and $\mathcal{Q}_i^d(\omega B, y')$ are interpolated linearly, since ωB is not necessarily part of the debt grid $\mathfrak{B}$.

- (e) Update expected value functions $\mathcal{F}$, default probability λ and pecuniary payoffs m w.r.t. the persistent exogenous state:

$$\begin{aligned} F_{i+1}(B, y) &= \Pi \mathcal{F}_{i+1}(B, y'), & F_{i+1}^d(B, y) &= \Pi \mathcal{F}_{i+1}^d(B, y'), \\ m_{i+1}^r(B, y) &= \Pi k_{i+1}^r(B, y'), & m_{i+1}^d(B, y) &= \Pi k_{i+1}^d(B, y'), \\ \lambda_{i+1}(B, y) &= \Pi \mathcal{D}_{i+1}(B, y'). \end{aligned}$$

- (f) Compute haircuts κ , haircut-weighted collateral Θ , and adjusted bond holdings for both investor types in the repayment case, and combine these elements to obtain the non-pecuniary part of the payoff, Λ :

$$\begin{aligned} \kappa_{i+1}^r(B, y) &= \kappa(\lambda_{i+1}), \\ \Theta_{i+1}^r(B, y) &= (1 - \kappa_{i+1}^r(B, y)) \times m_{i+1}^r(B, y') \times B, \\ \Theta_{i+1}^d(B, y) &= (1 - \bar{\kappa}) \times m_{i+1}^d(B, y') \times B, \\ \Theta_{H,i+1}^r(B, y) &= \Theta_{i+1}^r(B, y) - \frac{1}{2} \zeta_2, \\ \Theta_{H,i+1}^d(B, y) &= \Theta_{i+1}^d(B, y) - \frac{1}{2} \zeta_2, \\ \Theta_{L,i+1}^r(B, y) &= \Theta_{i+1}^r(B, y) + \frac{1}{2} \zeta_2, \\ \Theta_{L,i+1}^d(B, y) &= \Theta_{i+1}^d(B, y) + \frac{1}{2} \zeta_2, \\ \Lambda_{i+1}^r(B, y) &= (1 - \kappa_{i+1}^r(B, y)) \left(\frac{1}{2} \left(\frac{1}{2} u'(\Theta_{L,i+1}^r(B, y')) - \frac{1}{2} u'(\Theta_{i+1}^r(B, y')) \right) \right. \\ &\quad \left. + \frac{1}{2} \left(\frac{1}{2} u'(\Theta_{H,i+1}^r(B, y')) - \frac{1}{2} u'(\Theta_{i+1}^r(B, y')) \right) \right) \\ \Lambda_{i+1}^d(B, y) &= (1 - \bar{\kappa}) \left(\frac{1}{2} \left(\frac{1}{2} u'(\Theta_{L,i+1}^d(B, y')) - \frac{1}{2} u'(\Theta_{i+1}^d(B, y')) \right) \right. \\ &\quad \left. + \frac{1}{2} \left(\frac{1}{2} u'(\Theta_{H,i+1}^d(B, y')) - \frac{1}{2} u'(\Theta_{i+1}^d(B, y')) \right) \right). \end{aligned}$$

- (g) Update bond prices in repayment and default :

$$\begin{aligned} \mathcal{Q}_{i+1}^r(B, y) &= m_{i+1}^r(B, y) (1 + \Lambda_{i+1}^r(B, y)) / (1 + r^{rf}), \\ \mathcal{Q}_{i+1}^d(B, y) &= m_{i+1}^d(B, y) (1 + \Lambda_{i+1}^d(B, y)) / (1 + r^{rf}). \end{aligned}$$

- (h) Apply the cap on bond prices.

2. Compute convergence criteria:

$$\Delta \mathcal{F}_{t+1} = \max \left\{ \left\| \mathcal{F}_{t+1} - \mathcal{F}_t \right\|_{\infty}, \left\| \mathcal{F}_{t+1}^d - \mathcal{F}_t^d \right\|_{\infty} \right\}$$

$$\Delta \mathcal{Q}_{t+1} = \max \left\{ \left\| \mathcal{Q}_{t+1}^r - \mathcal{Q}_t^r \right\|_{\infty}, \left\| \mathcal{Q}_{t+1}^d - \mathcal{Q}_t^d \right\|_{\infty} \right\}$$

3. If $\Delta \mathcal{Q}_{t+1} < \varepsilon \wedge \Delta \mathcal{W}_{t+1} < \varepsilon$, stop. Otherwise, set $t = t + 1$ and go to step 1.

Table C.1 summarises the parameters governing the numerical approximation. The results do not visibly change when the grid sizes are increased further. The standard deviation of the taste shock $\sigma_z = 0.5$ is chosen to ensure stable convergence for all parameter combinations, which is typically achieved in less than 1000 iterations.

The maximum default probability is set to 0.75, following Chatterjee and Eyigungor (2015). To rule out implausibly high valuations of collateral services for small amounts of outstanding debt, we also impose a minimum bond spread (or, alternatively, maximum convenience yield). The minimum spread of a five-year German government bond over the five-year interest-rate-swap rate observed in our sample (-105 basis points) serves as a lower bound on bond spreads in our calibration. Like the maximum default probability, this constraint does not bind during model simulation and all of our numerical results are robust to changes to either constraint.

Convergence criterion	$\varepsilon = 1\text{E-}05$
Number of grid points for income	$n_y = 101$
Size of income grid	$y \in [-3\sigma_y, +3\sigma_y]$
Number of grid points for debt	$n_B = 301$
Size of debt grid	$B \in [0, 6]$
Taste shock parameter	$\sigma_z = 0.5$
Maximum default probability	$\bar{\lambda} = 0.75$
Minimum bond spread	$\underline{s} = -105$ basis points

Table C.1: Parameters of the computational algorithm

D Additional quantitative results

This section contains additional quantitative results for different model versions considered in Section 6.1 and provides details on the various robustness checks and model extensions introduced in Section 6.3.

D.1 Supplementary calibration details and comparative statics

This section provides additional details on the different model versions analysed in Section 6.1.

Model fit in alternative model versions. Table D.1 displays the model fit of the three model variations with constant u' , constant convenience yield, and without convenience yield, respectively. In all cases, the debt-to-GDP ratio and CDS spread are matched well. Since the model with constant u' and constant convenience yield have one parameter less than the baseline model with debt-elastic collateral valuation, it is not possible to simultaneously fit the mean convenience yield and its volatility. Consequently, the model without convenience yield has two targeted moments less. As a by-product, the model without convenience yield also cannot simultaneously match the CDS spread, bond spread over the risk-free rate s_t^{rf} and over German Bunds s_t^{DE} .

	Constant u'		Constant CY		No CY	
<i>Mean</i>	Data	Model	Data	Model	Data	Model
$Q_t B_t / y_t$	4.49	4.47	4.49	4.44	4.49	4.49
cds_t	79	76	79	81	79	79
cy_t	32	29	32	30	32	0
s_t^{rf}	47	48	47	51	47	79
s_t^{DE}	100	72	100	81	100	79
$basis_t^{DE}$	-21	0	-21	0	-21	0
<i>Volatility</i>						
$Q_t B_t / y_t$	0.26	0.30	0.26	0.31	0.26	0.30
cds_t	113	176	113	182	113	177
cy_t	31	3	31	6	31	0
s_t^{rf}	115	172	115	176	115	177
s_t^{DE}	145	172	145	176	145	177
$basis_t^{DE}$	38	3	38	6	38	0

Table D.1: Model fit in alternative model versions. This table displays targeted and key untargeted moments for alternative specifications of convenience yield. We set $\zeta_1 = 0.0007$, $\beta = 0.964$ and $d_0 = 21.9141$ in the model with constant u' (left columns). We set $\zeta_1 = 0.0007$, $\mu = 0$, $\beta = 0.965$ and $d_0 = 21.5625$ in the model with constant convenience yield (middle columns). We set $\zeta_1 = 0$, $\beta = 0.962$ and $d_0 = 24.375$ in the model without convenience yield (right columns). Spreads are annualised and given in basis points, y_t refers to quarterly real GDP and $Q_t B_t$ is the market value of government bonds outstanding and $Q_t(B_{t+1} - B_t)$ is net debt issuance, at market values. Crisis episodes are all periods with a CDS spread exceeding 40bps. Targeted moments are shown in colour. All model-implied statistics are based on simulations of 500,000 periods, excluding 5,000 burn-in periods. We also exclude all periods in which the government is in financial autarky as well as 40 quarters after reentering financial markets following an exclusion period.

Comparative statics. In Table D.2, we provide a pure comparative statics exercise to complement our model comparison in Table 4. To do so, we fix $\beta = 0.965$ and $d_0 = 21.5625$, which matches CDS spread and debt-to-GDP ratio in the model with constant convenience yield $\tilde{\Lambda}$ and is shown in column (3). Panel A of Table D.2 shows how the presence and the shape of convenience yield affects key welfare-relevant moments. Adding elastic haircuts in column (2) reduces the default rate, nominal debt outstanding B_t/y_t and collateral availability θ_t/y_t , which comes also with more periods with a negative bond spread. The baseline with collateral scarcity effects in column (1) features a much smaller share of negative spread periods and a default rate spiking from 1.17% to 2.58%, which is accompanied by a substantial increase in nominal debt outstanding. Lastly, the model without convenience yield in column (4) exhibits substantially less debt outstanding. Interestingly, the model without convenience yield is also characterised by a slightly elevated default rate. The reason is that a higher bond price associated with convenience yield make bond market access more valuable to the government. This channel is not present in the simplified model.

Panels B, C and D show that there is a similar time variation in sovereign risk as measured by CDS spreads, but a remarkable difference in the time variation of convenience yield. It is essentially constant over time in the models with constant u' , constant Λ and trivially absent in the model with $\zeta_1 = 0$. This has also effects on the correlation of the debt-to-GDP ratio with CDS spreads and the relative funding advantage vis-à-vis German Bunds, as shown in Panel E. In addition, Panel E also illustrates that public debt management is more countercyclical if marginal convenience yield is positive, compared to the model with constant convenience yield and, in particular, negative marginal convenience yield, i.e. column (2) with a constant u' . The reason behind this is that bond prices depreciate less if the government takes on more default risk: While the pure default risk compensation still induces a decline in bond prices, the positive marginal convenience yield dampens this decline and enables more public consumption smoothing.

	(1) Elastic u' $\partial\Lambda/\partial b > 0$	(2) Constant u' ($\zeta_2 = 0$) $\partial\Lambda/\partial b < 0$	(3) Constant Λ ($\zeta_2 = \mu = 0$) $\partial\Lambda/\partial b = 0$	(4) No CY ($\zeta_1 = 0$) -
Marginal convenience yield				
<i>Panel A: Key moments</i>				
Negative spread (%)	6.9	64.4	58.1	0
Default rate (%)	2.58	1.17	1.42	1.95
ave(B_t/y_t)	4.28	3.91	3.93	3.72
ave(θ/y_t)	4.39	4.29	4.40	-
<i>Panel B: Financial markets</i>				
ave(cds_t)	152	68	81	110
ave(cy_t)	38	29	30	0
ave($basis_t^{DE}$)	-33	0	2	0
<i>Panel C: Safe periods</i>				
ave(cds_t)	26	17	18	22
ave(cy_t)	19	28	28	0
ave($basis_t^{DE}$)	-10	0	0	0
<i>Panel D: Crisis periods</i>				
ave(cds_t)	230	213	218	221
ave(cy_t)	50	30	33	0
ave($basis_t^{DE}$)	-47	2	5	0
<i>Panel E: Debt management and cyclical properties</i>				
ave($Q_t B_t/y_t$)	4.69	4.44	4.40	4.09
ave($Q_t(B_{t+1} - B_t)/y_t$)	0.013	0.006	0.007	0.007
ave($Q_t(B_{t+1} - B_t)/y_t$), crisis	0.027	0.028	0.028	0.027
cor($Q_t(B_{t+1} - B_t)/y_t, y_t$)	-0.81	-0.69	-0.71	-0.76
cor($Q_t B_{t+1}/y_t, cds_t$)	0.34	0.09	0.13	0.20
cor($Q_t B_{t+1}/y_t, basis_t^{DE}$)	-0.57	0.16	0	0

Table D.2: The role of convenience yield – comparative statics. Spreads are annualised and given in basis points, y_t refers to quarterly real GDP. $Q_t B_t$ is the market value of government bonds outstanding and $Q_t(B_{t+1} - B_t)$ is net debt issuance, at market values. Crisis episodes are all periods with a CDS spread exceeding 40bps. All model-implied statistics are based on simulations of 500,000 periods, excluding 5,000 burn-in periods. We also exclude all periods in which the government is in financial autarky as well as 40 quarters after reentering financial markets following an exclusion period. The (annualised) default rate and the share of periods with a negative bond spread is based on periods where the government has access to financial markets

D.2 Model extension: Market illiquidity

This section presents the investor problem with trading frictions and the derivation of a market illiquidity-augmented bond-pricing condition. Investor types are revealed in the first sub-period. Denote the investor type by $i \in \{L, H\}$ for low and high collateral valuation, i.e. $u_L(\cdot) < u_H(\cdot)$. Investors adjust their bond holdings in exchange for an endogenously determined fee $\phi_{i,t}$. Dealers have access to a competitive inter-dealer market and intermediate between buyers and sellers. The terms of trade are determined bilaterally between a dealer and an investor via Nash bargaining. The bargaining power of both sides

is normalised to $\frac{1}{2}$ in the following. We assume that there are no search frictions, such that each buyer/seller is matched to a dealer. In the second sub-period, investors receive a constant endowment e . Government revenue and market re-entry shocks realize at the beginning of the second sub-period, before the centralised government bond market opens. Figure D.1 summarises the within-period timing structure.

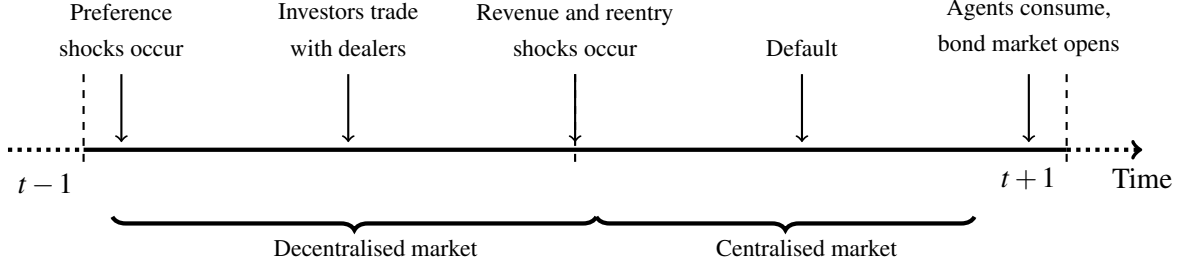


Figure D.1: Within-period timing

Maximisation problem. The investor-decision problem is formulated recursively. An investor's value function in the second sub-period (centralised market) is given as

$$\mathcal{W}_t(\tilde{b}_t, a_t) = \max_{b_t, c_t} \left\{ c_t + \frac{1}{1+r^f} \mathbb{E}_t [\mathcal{V}_{t+1}(b_t)] \right\}$$

subject to the budget constraint

$$c_t = e - a_t + k_t \tilde{b}_t - q_t b_t,$$

where e is a constant endowment, $\tilde{b}_t$ are bond holdings at the beginning of the second sub-period and a_t are payments owed to a dealer from trading in the decentralised market in the first sub-period of period t (described below).

The continuation value in the second sub-period reflects the occurrence of idiosyncratic preference shocks. The probability of becoming an L -type investor is normalised to $\frac{1}{2}$. An investor's value function at the beginning of a period, before idiosyncratic preference shocks materialise, is given by

$$\mathcal{V}_t(b_{t-1}) = \frac{1}{2} \mathcal{V}_{L,t}(b_{t-1}) + \frac{1}{2} \mathcal{V}_{H,t}(b_{t-1}),$$

with the value of being an i -type investor given by

$$\begin{aligned} \mathcal{V}_{i,t}(b_{t-1}) &= u_i((1 - \kappa_t) m_t \tilde{b}_{i,t}) + \mathbb{E}_{t-1} [\mathcal{W}_t(\tilde{b}_{i,t}, -\tilde{q}_t(\tilde{b}_{i,t} - b_{t-1}) - \phi_{i,t})] \\ &= u_i((1 - \kappa_t) m_t \tilde{b}_{i,t}) + m_t(\tilde{b}_{i,t} - b_t) - \tilde{q}_t(\tilde{b}_{i,t} - b_{t-1}) - \phi_{i,t} + \mathbb{E}_{t-1} [\mathcal{W}_t(b_t, 0)], \end{aligned}$$

which holds for an arbitrary b_t . Here, $\phi_{i,t}$ is a fee charged by dealers and $\tilde{q}_t$ is the competitive bond price on the inter-dealer market (both described below). It should be noted that the second equality is due to the linearity of the value function $\mathcal{W}_t(\cdot)$ in $\tilde{b}_t$ and a_t .

Decentralised market. To adjust their bond holdings in response to the i shock, investors contact dealers. The terms of trade between dealers and investors are determined bilaterally via Nash bargaining. For an i -type investor, the bargaining threat point is

$$\bar{V}_{i,t}(b_{t-1}) = u_i((1 - \kappa_t)m_t b_{t-1}) + \mathbb{E}_{t-1}[\mathcal{W}_t(b_{t-1}, 0)],$$

such that the surplus from trading is given by the utility gain from trading $u_i(\tilde{\theta}_{i,t}) - u_i((1 - \kappa_t)m_t b_{t-1})$, net of pecuniary benefits, and the fee paid to dealers

$$S_{i,t}(b_{t-1}) = u_i(\tilde{\theta}_{i,t}) - u_i((1 - \kappa_t)m_t b_{t-1}) + m_t(\tilde{b}_{i,t} - b_{t-1}) - \tilde{q}_t(\tilde{b}_{i,t} - b_{t-1}) - \phi_{i,t}.$$

In the expression for the surplus, $\tilde{\theta}_{i,t} \equiv (1 - \kappa_t)m_t \tilde{b}_{i,t}$ is the (haircut-adjusted) value of bond holdings after adjustment for the bond position and $(\tilde{b}_{i,t}, \phi_{i,t})$ constitute the terms of trade, which consist of the investor's bond holdings after the meeting $\tilde{b}_{i,t}$ and the fee charged by the dealer $\phi_{i,t}$. Payments owed to dealers consist of the fee and the desired adjustment of the bond positions,

$$a_{i,t} = \phi_{i,t} + \tilde{q}_t(\tilde{b}_{i,t} - b_{t-1}),$$

and are settled in the subsequent centralised market. A dealer's surplus in an i -type meeting simply equals $\phi_{i,t}$, which is consumed by the dealer in the centralised market. Dealers do not acquire bonds in the centralised market. The investors' bargaining power is α , the terms of trade solve the generalised Nash bargaining problem

$$\max_{\tilde{b}_{i,t}, \phi_{i,t}} \left[u_i((1 - \kappa_t)m_t \tilde{b}_{i,t}) - u_i((1 - \kappa_t)m_t b_{t-1}) + m_t(\tilde{b}_{i,t} - b_{t-1}) - \tilde{q}_t(\tilde{b}_{i,t} - b_{t-1}) - \phi_{i,t} \right]^\alpha \phi_{i,t}^{1-\alpha},$$

which leads to the two first-order conditions

$$\tilde{q}_t = (1 - \kappa_t)m_t u'_i(\theta_{i,t}) + m_t, \tag{D.1}$$

$$\phi_{i,t} = (1 - \alpha) \left(u_i(\tilde{\theta}_{i,t}) - u_i(\theta_t) + m_t(\tilde{b}_{i,t} - b_{t-1}) - \tilde{q}_t(\tilde{b}_{i,t} - b_{t-1}) \right). \tag{D.2}$$

It should be noted that the dealer fee simply equals the dealer's bargaining power $1 - \alpha$ multiplied by the total surplus $S_{i,t}(b_{t-1}) + \phi_{i,t}$. As in [Lagos and Rocheteau \(2009\)](#), the dealer fee can be expressed in terms of a meeting-specific bond price $\tilde{q}_{i,t}$:

$$\phi_{i,t} = (\tilde{q}_{i,t} - \tilde{q}_t)(\tilde{b}_{i,t} - b_{t-1}).$$

Using this relationship, the meeting-specific bond price can be expressed as a weighted average of the inter-dealer market price $\tilde{q}_t$ and the total surplus net of payments $\tilde{q}_{i,t}(\tilde{b}_{i,t} - b_{t-1})$ divided by the net trading position:

$$\tilde{q}_{i,t} = \alpha \tilde{q}_t + (1 - \alpha) \frac{u_i(\tilde{\theta}_{i,t}) - u_i(\theta_t) + m_t(\tilde{b}_{i,t} - b_{t-1})}{\tilde{b}_{i,t} - b_{t-1}}.$$

This will be useful for deriving bid-ask spreads below. If the investor holds all bargaining power ($\alpha = 1$), the dealer does not charge a mark-up/mark-down, such that $\tilde{q}_{i,t} = \tilde{q}_t$. If the investor is a net buyer ($\tilde{b}_{i,t} > b_{t-1}$), then the ask price of the dealer is $\tilde{q}_{i,t}$, which exceeds the inter-dealer price $\tilde{q}_t$ whenever $\alpha < 1$. Similarly, if the investor is a net seller, $\tilde{q}_{i,t}$ is the bid price, which is below $\tilde{q}_t$ for $\alpha < 1$. Consistent with the quantitative analysis, we normalise the bargaining power to $\alpha = \frac{1}{2}$ in the following. A commonly used measure of the extent to which a market is affected by trading frictions is the bid-ask spread, which in our model is given as

$$\tilde{q}_{H,t} - \tilde{q}_{L,t} = \frac{\phi_{H,t}}{\tilde{b}_{H,t} - b_{t-1}} - \frac{\phi_{L,t}}{\tilde{b}_{L,t} - b_{t-1}}. \quad (\text{D.3})$$

Market clearing in the competitive inter-dealer market requires

$$\frac{1}{2}\tilde{b}_{L,t} + \frac{1}{2}\tilde{b}_{H,t} = B_{t-1},$$

with government bond supply B_{t-1} , such that

$$u'_L((1 - \kappa_t)m_t\tilde{b}_{L,t}) = u'_H\left((1 - \kappa_t)m_t \times 2 \times \left(B_{t-1} - \frac{1}{2}\tilde{b}_{L,t}\right)\right),$$

determines $\tilde{b}_{L,t}$ and $\tilde{b}_{H,t}$ via market clearing.

Centralised market. For the centralised market, the investor first-order condition is

$$-q_t + \frac{1}{1 + r^{rf}} \frac{d\mathbb{E}_t[\mathcal{V}_{t+1}(b_t)]}{db_t} = 0. \quad (\text{D.4})$$

The value $\mathcal{V}_{t+1}(\cdot)$ can be written as

$$\mathcal{V}_{t+1}(b_t) = \mathbb{E}_i[u_i(\theta_t) + S_{i,t+1}(b_t)] + \mathbb{E}_t[\mathcal{W}_{t+1}(b_t, 0)],$$

such that

$$\begin{aligned} \frac{d\mathcal{V}_{t+1}(b_t)}{db_t} &= \mathbb{E}_i \left[(1 - \kappa_{t+1})m_{t+1}u'_i(\theta_{t+1}) + \frac{dS_{i,t+1}(b_t)}{db_t} \right] + \frac{\partial \mathbb{E}_t[\mathcal{W}_{t+1}(b_t, 0)]}{\partial b_t} \\ &= \mathbb{E}_i \left[(1 - \kappa_{t+1})m_{t+1}u'_i(\theta_{t+1}) + \frac{dS_{i,t+1}(b_t)}{db_t} \right] + m_{t+1}. \end{aligned}$$

The effect of individual bond holdings on trading frictions is given by

$$\begin{aligned} \frac{dS_{i,t+1}(b_t)}{db_t} &= (1 - \kappa_{t+1})m_{t+1} \frac{d\tilde{b}_i(b_t)}{db_t} u'_i(\tilde{\theta}_{i,t+1}) - (1 - \kappa_{t+1})m_{t+1}u'_i(\theta_{t+1}) \\ &\quad + m_{t+1} \left(\frac{d\tilde{b}_i(b_t)}{db_t} - 1 \right) - \tilde{q}_{t+1} \left(\frac{d\tilde{b}_i(b_t)}{db_t} - 1 \right) - \frac{d\phi_i(b_t)}{db_t}. \end{aligned}$$

Using

$$\frac{d\phi_i(b_t)}{db_t} = \frac{1}{2} \left[(1 - \kappa_{t+1})m_{t+1} \frac{d\tilde{b}_i(b_t)}{db_t} u'_i(\tilde{\theta}_{i,t+1}) - (1 - \kappa_{t+1})m_{t+1} u'_i(\theta_{t+1}) \right. \\ \left. + m_{t+1} \left(\frac{db_i(b_t)}{db_t} - 1 \right) - \tilde{q}_{t+1} \left(d \frac{\tilde{b}_i(b_t)}{db_t} - 1 \right) \right],$$

together with (D.1), this derivative can be written as

$$\frac{dS_{i,t+1}(b_t)}{db_t} = \frac{1}{2} \left[u'_i(\tilde{\theta}_{i,t+1}) - u'_i(\theta_{t+1}) \right] (1 - \kappa_{t+1})m_{t+1},$$

such that

$$\frac{dV_{t+1}(b_t)}{db_t} = \mathbb{E}_i \left[u'_i(\theta_{t+1}) + \frac{1}{2} \times \left\{ u'_i(\tilde{\theta}_{i,t+1}) - u'_i(\theta_{t+1}) \right\} \right] (1 - \kappa_{t+1})m_{t+1} + m_{t+1}.$$

Combining this condition with the investor first-order condition (D.4) then yields the bond-pricing equation:

$$q_t = \frac{1}{1 + r^{rf}} \times \underbrace{m_{t+1}}_{\text{payoff}} \times \underbrace{(1 + \Lambda_{t+1})}_{\text{convenience yield}}, \quad (\text{D.5})$$

As in Lagos and Wright (2005), the linearity of investor preferences with respect to consumption and idiosyncratic preference shocks being i.i.d. implies that investors choose the same bond holdings b_t in the centralised market regardless of their current holdings $\tilde{b}_{i,t-1}$. Finally, the bid-ask spread as a measure of illiquidity is given by

$$b_a^h(B, y) = \frac{1}{2} \left[\frac{u_H(\Theta_H^h(B, y, \tilde{b}_H(B))) - u_H(\Theta^h(B, y, B))}{\tilde{b}_H(B) - B} - \frac{u_L(\Theta_L^h(B, y, \tilde{b}_L(B))) - u_L(\Theta^h(B, y, B))}{\tilde{b}_L(B) - B} \right],$$

which in turn depends on the credit status $h \in \{0, 1\}$.

Parameter choices We specify the period utility function of high-valuation and low-valuation investors as follows:

$$u_L(\theta_t) = -\frac{\zeta_1}{\zeta_2} \exp\{-\zeta_2(\theta_t + b^{alt})\}, \\ u_H(\theta_t) = -\frac{\zeta_1}{\zeta_2} \exp\{-\zeta_2(\theta_t + b^{alt} - \zeta_3 - \zeta_5 \mathbf{1}\{\log(y_t) < 0\})\}.$$

It follows that the wedge between bond and CDS prices is given by

$$\Lambda_t = (1 - \kappa_t) \zeta_1 \exp\{-\zeta_2((1 - \kappa_t)m_t B_{t-1} + b^{alt})\}.$$

The optimal bond holdings of high- and low-valuation investor types are given by

$$\begin{aligned}\theta_{L,t} &= (1 - \kappa_t)m_t B_{t-1} - \frac{1}{2}(\zeta_3 + \zeta_5 \mathbf{1}\{\log(y_t) < 0\}) , \\ \theta_{H,t} &= (1 - \kappa_t)m_t B_{t-1} + \frac{1}{2}(\zeta_3 + \zeta_5 \mathbf{1}\{\log(y_t) < 0\}) .\end{aligned}$$

We can use the parameter ζ_3 to tune the gains from trade between high-valuation and low-valuation investors. Note that this model of market illiquidity does not generate time-variation in bond trading volumes. Therefore, we add the parameter ζ_5 that allows us to generate the empirically observed uptake in bid-ask spreads during sovereign debt crises in a parsimonious way. For a micro-founded model that is capable of explaining this feature, we refer to [Chaumont \(2024\)](#) and [Passadore and Xu \(2022\)](#). It is natural to target the averages and volatilities of bid-ask spreads when setting $\zeta_3 = 1\text{E-}5$ and $\zeta_5 = 1\text{E-}3$. Consistent with the bond maturity in our calibration, the data moment for bid-ask spreads is computed using the on-the-run five-year-bond. The model fit is shown in column (2) of Table [D.3](#).

D.3 Model extension: Investor risk aversion

In contrast to the baseline model, this extension allows for risk-averse investors. Specifically, investors have preferences given by $\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \frac{w(c_t) + u(\theta_t)}{(1+r^f)^t} \right]$, where $w(\cdot)$ is an increasing and concave utility function over consumption c_t . The exact functional form is not crucial, as long as it is parameterized to a match an additional data moment related to term premia, such as the level or volatility of bond *yields* in addition to bond spreads that we have targeted in the baseline model.

Each period, investors receive an exogenous time-invariant endowment e . The first-order condition associated with the investor problem yields the bond pricing equation

$$q_t = \frac{1}{1+r^f} \frac{\tilde{m}_{t+1} + m_{t+1}\Lambda_{t+1}}{w'(c_t)} . \quad (\text{D.6})$$

The term $\tilde{m}_t = \mathbb{E}_{t-1} [w'(c_t)k_t]$ is the expected value of the future payoff weighted with the future marginal utility of investor consumption. As in the baseline model, the payoff k_t is given by equation [\(19\)](#). Since investor consumption,

$$c_t = e + b_{t-1}k_t - q_t b_t, \quad (\text{D.7})$$

depends on the bond price q_t and, at the same time, enters the bond pricing equation [\(D.6\)](#), we cannot formulate a bond pricing schedule only as a function of end-of-period debt and current output. Investor consumption becomes an additional constraint for the government and affects its policy problem.

In recursive notation, the bond price in the repayment case now satisfies

$$q(B, B', y) = \frac{1}{1+r^f} \left[\frac{\tilde{m}^r(B', y)}{w'(c(B, B', y))} + \frac{m^r(B', y)}{w'(e)} \Lambda^r(B', y) \right] , \quad (\text{D.8})$$

with

$$c(B, B', y) = e + (\tilde{\delta} + (1 - \delta)q(B, B', y))B - q(B, B', y)B', \quad (\text{D.9})$$

after imposing bond market clearing $b = B$.

The new function $\tilde{m}^r(B', y)$ is given as

$$\tilde{m}^r(B', y) = \mathbb{E}_{y'|y} \left[\begin{array}{l} (1 - \mathcal{D}(B', y'))(\tilde{\delta} + (1 - \delta)\mathcal{Q}^r(B', y'))w'(C^r(B', y')) \\ + \mathcal{D}(B', y')\mathcal{Q}^d(B', y')w'(e) \end{array} \right]$$

where $C^r(\cdot)$ is the policy function for equilibrium investor consumption in repayment. In the default (and autarky) case, investors simply consume their endowments, i.e. $c = e$, since debt payment is suspended and no new debt issuance is absorbed ($B' = B$). Importantly, investor consumption in default thus does not depend on the bond price. As a result, the equilibrium bond price in the default case is now given as

$$\mathcal{Q}^d(B, y) = \frac{1}{1 + r^{rf}} \left[\frac{\tilde{m}^d(B, y)}{w'(e)} + \frac{m^d(B, y)}{w'(e)} \Lambda^d(B, y) \right],$$

with expected marginal-utility weighted pecuniary benefits given by

$$\tilde{m}^d(B', y) = \vartheta \omega \tilde{m}^r(\omega B', y) + (1 - \vartheta) \mathbb{E}_{y'|y} \left[\mathcal{Q}^d(B', y') w'(e) \right],$$

and future non-pecuniary ones given by

$$\Lambda^d(B', y) = (1 - \kappa(1, 0)) \times u'(\Theta^d(B', y)),$$

Risk-averse bond pricing substantially complicates the numerical solution, as current investor consumption directly enters the bond pricing condition (D.6), but itself also depends on the current bond price – in addition to beginning-of-period and end-of-period bond holdings. To address this issue, within each iteration, we jointly solve for $c(B, B', y)$ and $q(B, B', y)$ using the two conditions (D.8) and (D.9) for every combination of y , B and B' , and then use the resulting bond pricing schedule, now a three-dimensional object, as an input for the government's borrowing problem. Note that investor risk aversion also affects the pricing of CDS spreads, the German Bund, and of a synthetic risk-free asset reflecting interest rate swaps.

Parameter choices In line with our functional form assumptions on convenience yield, we set $w(c_t) = -\frac{\xi_1}{\xi_2} \exp\{-\xi_2 c_t\}$ and then have $w'(c_t) = \xi_1 \exp\{-\xi_2 c_t\}$. We normalize investors' endowment to $e = 1$. Furthermore, we set $\xi_1 = 1$, as the bond price schedule is governed by the ratio of current and future marginal utilities, such that ξ_1 cancels out. The CARA parameter is set to $\xi_2 = 0.01$, which matches the average the government bond yield, which corresponds to 178 basis points in the data. Due to the absence of risk premia, the baseline model implies an average bond yield of 155 basis points. We slightly increase $d_0 = 21$ in order to match the CDS spread, which would otherwise exceed its data moment, as it is now also priced by risk-averse investors. The model fit is presented in column (3) of Table D.3.

D.4 Model extension: Time-varying collateral valuation

In the extension with time-varying collateral valuation, we modify investors' period utility as follows:

$$u(\theta_t) = -\frac{\zeta_1 + \zeta_4 \mathbf{1}\{\log(y_t) < 0\}}{\zeta_2} \exp\{-\zeta_2(\theta_t + b^{alt})\}.$$

The new parameter ζ_4 reflects the idea that collateral is particularly valuable to investors in crisis times, i.e. when income is below trend ($\log(y_t) < 0$). To discipline the additional free parameter, we use the government bond spread in crisis episodes as additional targeted moment. Our baseline calibration implies a slight over-reaction of the government bond spread (164 basis points) compared to the data (153 basis points, see Table 2). By assuming a higher collateral demand in times of crisis, which is obtained from setting $\zeta_4 = 400$, the bond spread increases to 150 basis points in crisis times. We recalibrate the weighting parameter to $\zeta_1 = 160$ to ensure an average convenience yield of 30 basis points. The model fit is summarised in column (4) of Table D.3.

D.5 Model extension: Time-varying alternative collateral supply

To incorporate a time-varying supply of alternative collateral into our model, we slightly modify the period utility of investors:

$$u(\theta_t) = -\frac{\zeta_1}{\zeta_2} \exp\{-\zeta_2(\theta_t + b_t^{alt})\}.$$

Here, alternative collateral b_t^{alt} depends on the sovereign endowment $b_t^{alt} = \zeta_6 \log(y_t) + b^{alt}$, where the new parameter is set to $\zeta_6 = -11.5$. This ensures that alternative collateral supply is given by $b_t^{alt} = 5.25$, which is the maximum value of German debt outstanding relative to Italian GDP over our sample (2012Q2), if the government's income satisfies $\log(y_t) = -0.0789$, which is the lowest point on the income grid. Consequently, investors collateral valuation of Italian government debt is particularly low if Italy's public finances are under stress. By contrast, the supply of alternative collateral declines to 3.43 if the endowment is at its maximum. This value is quite similar to the minimum amount of German debt outstanding (relative to Italian GDP), which was 3.69 in 2001Q1. Note that we have taken a conservative approach to parameterise ζ_6 , since German and Italian business cycles are not perfectly correlated and German fiscal policy also does not respond immediately to its own business cycle. The model fit is summarised in column (5) of Table D.3.

Table D.3 summarizes the fit of each model extension. The first column shows that evaluating the haircut function at the one-year ahead default probability yields almost the same fit to the data as the baseline model. Similarly, for the extension with market illiquidity all moments are remarkably close to the baseline model, see Column (2). This suggests that market illiquidity does not materially affect the interactions between convenience yield, default risk, and public debt management. Importantly, both the level and the time-series volatility of bid-ask spreads match their data counterparts reasonably well. Column (3) corresponds to the case with time-varying convenience yield. By increasing convenience yield during crisis periods, spreads are generally less volatile than in the baseline model, in particular the volatility of the spread over the German Bund is closer to its data moment of 145bps. The fit of the extension with time-varying alternative collateral supply in Column (4) is again similar to the baseline

	(1) 1y-ahead λ		(2) Illiquidity		(3) Time-var. u'		(4) Time-var. b^{alt}		(5) Risk aversion	
<i>Mean</i>	Data	Model	Data	Model	Data	Model	Data	Model	Data	Model
$Q_t B_t / y_t$	4.49	4.50	4.49	4.47	4.49	4.55	4.49	4.49	4.49	4.53
cds_t	79	80	79	75	79	76	79	80	79	76
cy_t	32	33	32	31	32	33	32	28	32	30
ba_t	-	-	11	9	-	-	-	-	-	-
$s_t^{rf} crisis_t$	-	-	-	-	153	155	-	-	-	-
r_t	-	-	-	-	-	-	-	-	178	177
s_t^{rf}	47	47	47	44	47	44	47	51	47	47
s_t^{DE}	100	100	100	92	100	101	100	84	100	94
$basis_t^{DE}$	-21	-20	-21	-18	-21	-24	-21	-4	-21	-18
<i>Volatility</i>										
$Q_t B_t / y_t$	0.26	0.38	0.26	0.38	0.26	0.39	0.26	0.32	0.26	0.38
cds_t	113	179	113	174	113	174	113	180	113	178
cy_t	31	37	31	35	31	41	31	12	31	36
ba_t	-	-	16	37	-	-	-	-	-	-
s_t^{rf}	115	143	115	141	115	134	115	174	115	144
s_t^{DE}	145	201	145	194	145	134	145	185	145	199
$basis_t^{DE}$	38	27	38	24	38	32	38	6	38	25

Table D.3: Model fit for extensions. Spreads are annualised and given in basis points, y_t refers to quarterly real GDP. $Q_t B_t$ is the market value of government bonds outstanding and $Q_t (B_{t+1} - B_t)$ is net debt issuance, at market values. Crisis episodes are all periods with a CDS spread larger than 40bps. Targeted moments are shown in colour. All model-implied statistics are based on simulations of 500,000 periods, excluding 5,000 burn-in periods. We also exclude all periods in which the government is in financial autarky as well as 40 quarters after reentering financial markets following an exclusion period.

model, although the (untargeted) CDS-bond basis over the German bund is considerably smaller than in the baseline model. The reason is that collateral is generally more abundant when it is in high demand during fiscal crises, i.e. there are virtually no periods where collateral valuation *and* haircuts on Italian government bonds are high at the same time, see also Panel D of Table D.4. Column (5) refers to the case of risk-averse investors. Here, we demonstrate that the model fits the average government bond *yield* r_t very well, as it now contains an additional risk premium component that amounts to around 25bps, which is slightly smaller than the average convenience yield. Notably, the other targeted moments are still matched well. Untargeted volatilities are slightly larger, but still quite similar to the baseline model.

In Table D.4, we explore key welfare-relevant moments and important financial market variables such as default risk, convenience yield, and the basis over the German Bund in the full sample and during safe and crisis periods, respectively. The respective spreads s_t^{rf} and s_t^{DE} can be backed out immediately from the CDS spread, the convenience yield and the basis over the Bund. While all extensions exhibit and overall similar patterns, in particular for the market illiquidity extension, some differences are noteworthy. The basis is much more responsive during crisis times in the case of time-varying collateral valuation u' , as it declines to -61bps, compared to -45bps in the baseline, but CDS spread and default are quite comparable to the baseline. By contrast, the default rate is somewhat larger in the case of risk-averse investors, while there negligible differences in terms of the $basis_t^{DE}$ and its time variation.

	(1) 1y-ahead	(2) Illiquid.	(3) T.-var. u'_t	(4) T.-var. b_t^{alt}	(5) Risk av.	(6) Baseline
<i>Panel A: Key moments</i>						
Negative spread (%)	49.9	49.9	48.0	51.0	47.9	49.9
Default rate (%)	1.33	1.33	1.34	1.44	1.45	1.33
$\text{ave}(B_t/y_t)$	3.95	3.95	4.02	3.98	4.05	3.95
$\text{ave}(Q_t B_t/y_t)$	4.47	4.47	4.55	4.49	4.53	4.47
$\text{ave}(\theta/y_t)$	4.30	4.30	4.37	4.31	4.34	4.30
<i>Panel B: Financial markets</i>						
$\text{ave}(cds_t)$	80	75	76	80	76	75
$\text{ave}(cy_t)$	33	31	33	28	30	31
$\text{ave}(basis_t^{DE})$	-20	-18	-24	-4	-18	-18
<i>Panel C: Safe periods</i>						
$\text{ave}(cds_t)$	16	16	16	17	16	16
$\text{ave}(cy_t)$	23	22	21	30	20	22
$\text{ave}(basis_t^{DE})$	-7	-7	-5	-1	-6	-6
<i>Panel D: Crisis periods</i>						
$\text{ave}(cds_t)$	220	217	214	221	228	217
$\text{ave}(cy_t)$	54	52	59	26	53	52
$\text{ave}(basis_t^{DE})$	-50	-45	-61	-9	-47	-45
<i>Panel E: Debt management and cyclical properties</i>						
$\text{ave}(Q_t(B_{t+1} - B_t)/y_t)$	0.009	0.008	0.008	0.008	0.008	0.008
$\text{ave}(Q_t(B_{t+1} - B_t)/y_t crisis_t)$	0.032	0.032	0.031	0.031	0.032	0.032
$\text{cor}(Q_t(B_{t+1} - B_t)/y_t, y_t)$	-0.78	-0.78	-0.78	-0.74	-0.78	-0.78
$\text{cor}(Q_t B_{t+1}/y_t, cds_t)$	0.33	0.31	0.32	0.17	0.28	0.31
$\text{cor}(Q_t B_{t+1}/y_t, basis_t^{DE})$	-0.58	-0.57	-0.59	-0.39	-0.54	-0.57

Table D.4: The role of convenience yield in different model extensions. Spreads are annualised and given in basis points, y_t refers to quarterly real GDP. $Q_t B_t$ is the market value of government bonds outstanding and $Q_t(B_{t+1} - B_t)$ is net debt issuance, at market values. Crisis episodes are all periods with a CDS spread exceeding 40bps. Targeted moments are shown in colour. All model-implied statistics are based on simulations of 500,000 periods, excluding 5,000 burn-in periods. We also exclude all periods in which the government is in financial autarky as well as 40 quarters after reentering financial markets following an exclusion period. The (annualised) default rate and the share of periods with a negative bond spread is based on periods where the government has access to financial markets.

D.6 Calibration to Spain and Portugal

In this section, we demonstrate that our results are robust to calibrating our baseline model with a debt-elastic collateral valuation function to Spain and Portugal, respectively. For Portugal, we truncate the sample based on which data moments are computed in 2011Q1, the last quarter before Portugal received funds by the “Troika” of European Financial Stability Facility (EFSF), the European Commission, and the IMF. Re-estimating the parameters of the endowment process yields $\rho_y = 0.975$ and $\sigma_y = 0.0098$ for Spain and $\rho_y = 0.980$ and $\sigma_y = 0.0111$ for Portugal.

Due to the higher persistence of the endowment process, we reduce the minimum spending level $\underline{g}$ to ensure that it never exceeds income, which would imply negative public spending in autarky with

a positive probability. We furthermore recalibrate the government discount factor and the default cost parameter to $\beta = 0.975$ and $d_0 = 11$ for Spain, while they are set to $\beta = 0.975$ and $d_1 = 23.75$ for Portugal. Due to the considerably smaller debt-to-GDP ratio, we recalibrate the parameters of the collateral valuation function to $\zeta_1 = 2000$ and $\zeta_2 = 1.5$ for Spain, and to $\zeta_1 = 4000$ and $\zeta_2 = 2.5$ for Portugal. In each case, the fit to financial market moments is shown in Table D.5. While the model is able to match targeted first and second moments and volatilities well, the average basis vis-à-vis Germany and its volatility are slightly smaller than in the data.

Table D.5: Model fit - calibration to Spain and Portugal

	Spain		Portugal	
<i>Mean</i>	Data	Model	Data	Model
$Q_t B_t / y_t$	2.00	1.87	3.12	3.03
$c d s_t$	78	91	60	45
$c y_t$	33	38	33	14
s_t^{rf}	44	53	23	31
s_t^{DE}	98	101	66	50
$basis_t^{DE}$	-20	-1	-11	-4
<i>Volatility</i>				
$Q_t B_t / y_t$	0.47	0.62	0.83	1.08
$c d s_t$	113	165	105	124
$c y_t$	37	26	23	20
s_t^{rf}	123	167	94	124
s_t^{DE}	150	172	111	128
$basis_t^{DE}$	45	14	20	12

Notes: Spreads are annualised and given in basis points, y_t refers to quarterly real GDP. $Q_t B_t$ is the market value of government bonds outstanding and $Q_t (B_{t+1} - B_t)$ is net debt issuance, at market values. Targeted moments are shown in colour. All model-implied statistics are based on simulations of 500,000 periods, excluding 5,000 burn-in periods. We also exclude all periods in which the government is in financial autarky as well as 40 quarters after reentering financial markets following an exclusion period.